A general framework for discretely sampled realized variance derivatives in stochastic volatility models with jumps

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Abstract

After the recent financial crisis, the market for volatility derivatives has expanded rapidly to meet the demand from investors, risk managers and speculators seeking diversification of the volatility risk. In this paper, we develop a novel and efficient transform-based method to price swaps and options related to discretely-sampled realized variance under a general class of stochastic volatility models with jumps. We utilize frame duality and density projection method combined with a novel continuous-time Markov chain (CTMC) weak approximation scheme of the underlying variance process. Contracts considered include discrete variance swaps, discrete variance options, and discrete volatility options. Models considered include several popular stochastic volatility models with a general jump size distribution: Heston, Scott, Hull-White, Stein-Stein, α-Hypergeometric, 3/2 and 4/2 models. Our framework encompasses and extends the current literature on discretely sampled volatility derivatives, and provides highly efficient and accurate valuation methods. Numerical experiments confirm our findings.

Keywords: Finance; volatility derivatives; regime-switching; jump diffusion; stochastic volatility

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1 Introduction

After the recent financial crisis, volatility risk, i.e. the uncertainty of asset volatility, has become a significant source of financial market risk and factors in as an important component in investment decisions and risk management. There is growing trading activity in volatility derivatives to either hedge against the volatility risk or speculate on volatility movements. According to a recent United States Commodity Futures Trading Commission (CFTC) report (Mixon and Onur (2015)) on the activity of over-the-counter (OTC) volatility derivatives, the total new swaps quoted in USD Vega was 159.4 million in May 2014 across several large exchanges. Here the quote is consistent with industry convention of "dollars per volatility point", or "dollar Vega". They also report that the total variance swaps outstanding as of May 2014 were measured at 2,164 millon USD Vega.

With the growing trading activity, volatility derivatives have attracted a significant amount of research activities both in industry and academia. One prominent example is the variance swap, which is an OTC derivative contract that pays at a fixed maturity T the difference between a given fixed level (fixed leg) and a realized level of variance (i.e. average of squared daily log-returns) over the swap's life (floating leg). Variance swaps on stock indices are liquid and popular investment tools for diversifying portfolios and transferring volatility risk. Less standardized variance swaps can be linked to other types of underlying stocks such as currencies or commodities, which may be useful for hedging volatility risk exposure or for taking positions on future realized volatility. For example, Carr and Lee (2007) price options on realized variance and volatility by using variance swaps as pricing and hedging instruments. While existing volatility derivatives are based on realized variance computed from the discretely sampled log stock price, the price of continuously sampled contracts are often used as approximations. As noted by Jarrow et al. (2013), most academic studies (see, for example, Howison et al. (2004), Windcliff et al. (2006), Benth et al. (2007), Sepp (2008), and Broadie and Jain (2008)) focus on continuously sampled variance and volatility swaps. As shown in Broadie and Jain (2008) for Bate's model (see Drimus et al. (2016) for an enlarged class), the approximation error is $\mathcal{O}(1/M)$, where M is the number of monitoring dates. As a result, we observe large pricing errors when the continuous approximation is used (see Lian et al. (2014)).

For discretely sampled variance swaps, it is difficult to use the elegant and model-free approach of Dupire (1993) and Neuberger (1994), who independently proved that the fair strike for a continuously sampled variance swap on any underlying price process with continuous path is simply two units of the forward price of the log contract. The Dupire-Neuberger theory was recently extended by Carr et al. (2012) to the case when the underlying stock price is driven by a time-changed Lévy process, thus allowing jumps in the path of the underlying stock price. In the literature, there is also research on the design of variance swap contracts so that there exists model-free hedges, see Martin (2013), Alexander and Rauch (2016), Rauch and Alexander (2016), and more recently the "divergence swaps" proposed in Schneider and Trojani (2015b). For valuation of discretely sampled variance swaps under parametric stochastic volatility models, see Broadie and Jain (2008), Zhu and Lian (2011), Bernard and Cui (2014), Lian et al. (2014), and Alexander and Rauch (2016). For a survey of academic literature in volatility derivatives, please refer to Carr and Lee (2009) and references therein.

Another area that has attracted a significant amount of research is the empirical study of "variance risk premiums", i.e. the premium investors demand for accepting the risk due to the randomness of the future variance. The existence of the variance risk premium has already been extensively documented in the literature, see Bakshi and Kapadia (2003), Carr and Wu (2009), Todorov (2010) and references therein. The prices of discretely sampled variance swaps play an important role as a proxy for the market realized variance in empirical studies; see recent studies in Wang et al. (2013), Dotsis (2015), Li and Li (2015), Neumann et al. (2016) and Konstantinidi and Skiadopoulos (2016). The knowledge of variance swaps or other higher order moment swaps can help us extract model-independent information about the non-linear risk premium in the market, as documented in recent literature Neuberger (2012), Kozhan et al. (2013), Schneider and Trojani (2015c), Schneider (2015), Schneider and Trojani (2015a), Rauch and Alexander (2016) and Filipović et al. (2016). Also see Coqueret and Tavin (2016) for an investigation of model risk in the context of variance swaps.

Stochastic volatility (SV) models are popular for modeling asset returns among practitioners and academics because they allow better fits to real option price data in the implied volatility surfaces, and they explicitly incorporate the *leverage* effect pervasive in financial markets. There is a vast literature on SV models, with representative formulations including Hull-White (Hull and White (1987)), Heston (Heston (1993)), and the 3/2 SV model (Lewis (2000)). There are also recent interests in developing new SV models to capture the financial market asset returns in the post-crisis era: α-Hypergeometric model (Da Fonseca and Martini (2016)), 4/2 model (Grasselli (2016)) and Jacobi model (Ackerer et al. (2016)). A large class of multivariate SV model, named the Wishart stochastic volatility model, is considered in Da Fonseca (2016). For empirical calibration of the SV models, please refer to Date and Islyaev (2015), Mrázek et al. (2016) and the references therein. The joint modeling of SV with stochastic interest rates has also received growing attention (Grezlak and Oosterlee (2011), Recchioni and Sun (2016)). However, the current literature on volatility derivatives is mostly limited to SV models *without jumps*, with the exception of approximations in a few special cases such as the Bates model (see Broadie and Jain (2008); Sepp (2008, 2012); Zheng and Kwok (2014)). On the other hand, empirical studies reveal the existence of sudden rapid movements in intra-day quotes of stock prices, i.e. jumps during the trading period, as documented in Bates (1996), Carr and Wu (2003), Lee and Hannig (2010), etc. The recent financial crisis highlights that event risks such as a market crash may be significant, e.g. the *flash crash* on May 6th, 2010.

Our paper aims to fill this gap in the literature, and proposes a new method to value swaps and options on discretely sampled realized variance under general SV models with a jump component. By blending regime-switching models, Markov chain approximation techniques and Fourier methods, we develop a fast and accurate approach, which we refer to as the projection (PROJ) method[1], for pricing volatility derivatives in SV models with jumps. Given the growing interest and importance in empirical studies of "variance risk premiums" in finance, of which the valuation of discretely sampled variance swaps is central to model calibration, our proposed *unified* framework provides an efficient method for the valuation and calibration of discretely sampled volatility derivatives in a large class of models. In general, Fourier methods for exotic pricing, such as Feng and Linetsky (2008) and Fusai et al. (2016), require a closed-form expression for the underlying characteristic function. Similarly, the methods of Pun et al. (2015) and Lian et al. (2014), respectively, depend heavily on the closed-form characteristic function of the underlying process and of the integrated variance, which may not be available for SV models such as the 4/2 model or the α-Hypergeometric model (see equation (5) below). In contrast, while our approach resides in the Fourier domain, characteristic functions of the underlying process or integrated variance are not required. Similar to Zheng and Kwok (2014), our approach can handle jumps in the underlying, but does not require the joint moment generating function of the underlying and variance process as they do. In addition, our method is more versatile since it can handle volatility swaps as well as call/put options written on the discrete realized variance. Our paper makes the following main contributions:

1. To handle general SV models with jumps, we combine a Markov chain approximation for the variance process with a regime-switching approximation for the underlying which allows us to recast the problem in terms of a simplified regime-switching jump diffusion model. The approach provides a *unified* method for pricing variance derivatives in many well-known SV models with jumps, which include those of Heston, Hull-White, Scott, Stein-Stein, α-Hypergeometric model, the 3/2 model, and the 4/2 model with an arbitrary jump size distribution.

2. We develop a simple recursive framework, which enables Fourier pricing techniques. In particular, we devise an efficient and accurate pricing algorithm based on the recent frame projection (PROJ) method. The method applies to Lévy processes, regime-switching

[1] It was introduced for European options in Kirkby (2015) and has advantages over several existing transform-based methods, e.g. Carr and Madan (1999), Feng and Linetsky (2008), Fang and Oosterlee (2008) as highlighted in Staunton (2015)

Lévy processes and SV jump diffusions for general contracts written on realized variance or volatility.

3. We provide numerical results for variance derivatives under SV models with jumps. For many of these models, prices have never been reported in the literature. We verify the accuracy of our method against benchmarks based on Monte Carlo simulation, and cases for which prices exist in the literature. Given the accuracy of our method, these results can be used as reference benchmark values for future research.

The rest of this paper is structured as follows: We recall the definitions of swaps and options on discretely sampled realized variance in Section 2. Section 3 introduces a regime-switching framework for pricing contracts on realized variance in general stochastic volatility models. Our PROJ method for swaps and options on discretely sampled realized variance is detailed in Section 4. Numerical examples are provided in Section 6. Section 7 concludes the paper.

2 Discrete Variance and Volatility Derivatives

Consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{Q})$, where $\mathbb{Q}$ is the risk-neutral measure which is assumed to be known given a suitable market price of risk. Consider the asset price S_t on the finite trading horizon $[0, T]$ with M monitoring dates $0 = t_0 < t_1 < \ldots < t_M = T$. The discrete realized variance over $[0, T]$ is defined by

$$V_M := \frac{AF}{M} \sum_{m=1}^{M} \left(\log \frac{S_{t_m}}{S_{t_{m-1}}} \right)^2, \tag{1}$$

where AF is the annualization factor ($AF = 12$ for monthly monitoring, $AF = 52$ for weekly monitoring, or $AF = 252$ for daily monitoring). Assuming uniform sampling at a rate $\Delta_t = T/M$ over $[0, T]$, we have $AF = 1/\Delta_t$, so that $AF/M = 1/T$. The method described below also applies equally well to the case of relative returns (see Remark 4).

1. A discrete variance swap is a forward contract that exchanges the discrete realized variance with a fixed strike. The fair strike K_{Var} of the variance swap is determined such that the price of the contract is 0 at initiation. Its value is given by the risk-neutral expectation of V_M, i.e,

$$K_{Var} = \mathbb{E}[V_M]. \tag{2}$$

2. A discrete volatility swap is a contract which is similar to the discrete variance swap. Its fair strike is given by

$$K_{Vol} = \mathbb{E}[\sqrt{V_M}]. \tag{3}$$

3. The option on discrete realized variance gives the holder the right but not the obligation to exchange the discrete realized variance for a fixed strike. The price of an European call option with strike K is given by

$$C = \mathbb{E}[e^{-rT}(V_M - K)^+]. \tag{4}$$

3 Stochastic Volatility Framework

Hull-White (Hull and White (1987))	$dS_t = S_t\gamma dt + S_t\sqrt{v_t}dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = a_v v_t dt + \sigma_v v_t dW_t^2$	$a_v \in \mathbb{R}$ $\sigma_v, v_0 > 0$
Scott (Scott (1987))	$dS_t = S_t\gamma dt + S_t\exp(v_t)dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = \eta(\theta - v_t)dt + \sigma_v dW_t^2$	$\eta, \theta, v_0 \in \mathbb{R}$ $\sigma_v > 0$
Stein-Stein (Stein and Stein (1991))	$dS_t = S_t\gamma dt + S_t v_t dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = \eta(\theta - v_t)dt + \sigma_v dW_t^2$	$\eta, \theta \in \mathbb{R}$ $\sigma_v, v_0 > 0$
Heston (Heston (1993))	$dS_t = S_t\gamma dt + S_t\sqrt{v_t}dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = \eta(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^2$	$\eta, \theta \in \mathbb{R}$ $\sigma_v, v_0 > 0$
3/2 (Lewis (2000))	$dS_t = S_t\gamma dt + S_t/\sqrt{\widehat{v}_t}dW_t^1 + S_t(e^{J_t}-1)dN_t$ $d\widehat{v}_t = \widehat{\eta}[\widehat{\theta} - \widehat{v}_t]dt + \widehat{\sigma}_v\sqrt{\widehat{v}_t}dW_t^2, \quad \widehat{v}_0 = 1/v_0$	$\eta, \theta \in \mathbb{R}$ $\sigma_v, v_0 > 0$
4/2 (Grasselli (2016))	$dS_t = S_t\gamma dt + S_t[a_S\sqrt{v_t} + \frac{b_S}{\sqrt{v_t}}]dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = \eta(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^2$	$\eta, \theta, a_S, b_S \in \mathbb{R}$ $\sigma_v, v_0 > 0$
α-Hypergeometric (Da Fonseca and Martini (2016))	$dS_t = S_t\gamma dt + S_t\exp(v_t)dW_t^1 + S_t(e^{J_t}-1)dN_t$ $dv_t = (\eta - \theta\exp(a_v v_t))dt + \sigma_v dW_t^2$	$\eta, v_0 \in \mathbb{R}$ $\sigma_v, \theta, a_v > 0$

Table 1: SV dynamics for common models extended by jumps, with $\gamma := r - q - \lambda\kappa$. Note that for the Scott model and the α-Hypergeometric model, the underlying driver v_t is permitted to take negative values. The 3/2 dynamics are given for the reformulated process based on a change of variables: $\widehat{v}_t := 1/v_t, \widehat{\eta} := \eta\theta, \widehat{\theta} := (\eta + \sigma_v^2)/\widehat{\eta}, \widehat{\sigma}_v = -\sigma_v$. For the jump component, λ is the jump arrival rate and $\kappa := \mathbb{E}[e^{J_t} - 1]$.

Consider the following two-dimensional system of correlated time-homogeneous diffusions with independent jumps in the asset price dynamics[2]

$$\begin{cases} dS_t = \gamma S_t dt + m(v_t)S_t dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2, \end{cases} \tag{5}$$

where $\mathbb{E}[dW_t^1 dW_t^2] = \rho dt$ with $\rho \in (-1, 1)$. Here N_t is a Poisson process with jump rate λ, $J_t \sim J$ is the jump amplitude, and both are independent of W_t^1 and W_t^2. Table 1 provides the dynamics for a number of common SV models which are considered in our framework. The term $\gamma = r - q - \lambda\kappa$ captures the risk-neutral drift of the process compensated for jumps, where $\kappa = \mathbb{E}[e^J - 1]$, $r \geq 0$ is the common risk-free interest rate, and $q \geq 0$ is the continuous dividend yield.

[2]We note that our present methodology applies to the case of jump-free volatility dynamics, while the underlying itself may jump. A natural extension for future research is to allow jumps in volatility. Empirical evidence in some markets documents the improved fit for such models Bao et al. (2012).

Jump Type	$\nu(y)$	$\phi(\xi)$
Normal	$\frac{1}{\sqrt{2\pi}b_1}e^{-(x-a_1)^2/2b_1^2}$	$e^{\mathrm{i}\xi a_1-\frac{1}{2}\xi^2 b_1^2}$
DE	$p\eta_1 e^{-\eta_1 y}\mathbb{I}_{\{y\geq 0\}}+(1-p)\eta_2 e^{\eta_2 y}\mathbb{I}_{\{y<0\}}$	$p\frac{\eta_1}{\eta_1-\mathrm{i}\xi}+(1-p)\frac{\eta_2}{\eta_2+\mathrm{i}\xi}$
Mixed Normal	$p\frac{1}{\sqrt{2\pi}b_1}e^{-\frac{(y-a_1)^2}{2b_1^2}}+(1-p)\frac{1}{\sqrt{2\pi}b_2}e^{-\frac{(y-a_2)^2}{2b_2^2}}$	$pe^{\mathrm{i}\xi a_1-\frac{1}{2}\xi^2 b_1^2}+(1-p)e^{\mathrm{i}\xi a_2-\frac{1}{2}\xi^2 b_2^2}$

Table 2: Lévy measure $\nu(y)$ and characteristic function $\phi(\xi)$ of common jump distributions. With jump intensity λ, the characteristic exponent of the jump contribution is $\lambda(\phi(\xi)-1)$. Note that $\kappa := \phi(-i)-1$, and $\lambda\kappa$ is the drift compensator. DE: double exponential.

Let S_t^c denote the continuous part of the asset price. From Itô's formula for jump processes, it follows that

$$\begin{aligned} d\log(S_t) &= \frac{1}{S_t}dS_t^c - \frac{1}{2S_t^2}(dS_t^c)^2 + d\left[\sum_{k=1}^{N_t} Y_k\right] \\ &= \left(\gamma - \frac{1}{2}m^2(v_t)\right)dt + m(v_t)dW_t^1 + d\left[\sum_{k=1}^{N_t} Y_k\right]. \end{aligned} \tag{6}$$

Our modeling approach is to approximate the underlying variance process v_t by a locally consistent continuous-time Markov chain (CTMC), which reduces its state-space to a finite set. We then cast the model as a regime-switching jump diffusion, with regimes governed by the transition dynamics of the CTMC. By utilizing the characteristic function of the approximated process, we can conduct pricing in the Fourier space.

The first step is to reformulate (6) to separate $m(v_t)$ from dW_t^1. Hence we define

$$\hat{f}(x) := \int_c^x \frac{m(u)}{\hat{\sigma}(u)}du, \quad h(x) := \mathcal{L}(\hat{f}(x)) = \hat{\mu}(x)\hat{f}'(x) + \frac{1}{2}\hat{\sigma}^2(x)\hat{f}''(x),$$

where c is an arbitrary constant, and let $f(v_t, v_0) := \rho(\hat{f}(v_t) - \hat{f}(v_0))$. By an application of Itô's Lemma, the dynamics of $f(v_t, v_0)$ satisfy

$$df(v_t, v_0) = \rho d\hat{f}(v_t) = \rho h(v_t)dt + \rho m(v_t)dW_t^2. \tag{7}$$

We then define $W_t^* := \frac{W_t^1 - \rho W_t^2}{\sqrt{1-\rho^2}}$, which is a standard Brownian motion satisfying $\mathbb{E}[dW_t^* dW_t^2] = 0$, from which W_t^* and W_t^2 are independent. Substituting (7) into (6) yields

$$\begin{aligned} d\log(S_t) &= \left(\gamma - \frac{1}{2}m^2(v_t)\right)dt + m(v_t)(\rho dW_t^2 + \sqrt{1-\rho^2}dW_t^*) + d\left[\sum_{k=1}^{N_t} Y_k\right] \\ &= \left(\gamma - \frac{1}{2}m^2(v_t)\right)dt + df(v_t, v_0) - \rho h(v_t)dt + \sqrt{1-\rho^2}m(v_t)dW_t^* + d\left[\sum_{k=1}^{N_t} Y_k\right]. \end{aligned} \tag{8}$$

With $\widetilde{X}_t = \log(S_t/S_0) - f(v_t, v_0)$, we can rewrite (5) as

$$\begin{cases} d\widetilde{X}_t = \left(\gamma - \frac{1}{2}m^2(v_t) - \rho h(v_t)\right)dt + \sqrt{1-\rho^2}m(v_t)dW_t^* + d\left[\sum_{k=1}^{N_t} Y_k\right], \\ dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2. \end{cases} \tag{9}$$

The processes $\widetilde{X}_t$ and v_t are now driven by independent Brownian motions. The next step is to approximate the process v_t by a locally consistent CTMC.

3.1 Markov Chain Approximation of Variance Process

Recall that a finite state CTMC on $(\Omega, \mathcal{F}, \mathbb{Q})$ is a stochastic process $\{\alpha(t), t \geq 0\}$ which resides in a set $\mathcal{M} := \{1, 2, \ldots, m_0\}$ of possible states, and satisfies

$$\mathbb{Q}\left[\alpha(t+\Delta_t) = k | \alpha(t) = j, \alpha(t'), 0 \leq t' \leq t\right] = \mathbb{Q}[\alpha(t+\Delta_t) = k | \alpha(t) = j],$$

for all $j, k \in \mathcal{M}$ and $t, \Delta_t \geq 0$ and $0 \leq t' \leq t$. The transition dynamics of $\alpha(t)$ are described by the generator $Q = [q_{jk}]_{m_0 \times m_0}$, whose elements q_{jk} satisfy (i) $q_{jj} \leq 0$, and $q_{jk} \geq 0$, if $j \neq k$, and (ii) $\sum_k q_{jk} = 0$, $\forall j \in \mathcal{M}$. In terms of q_{jk}, the current regime makes transitions according to

$$\mathbb{Q}[\alpha(t+\Delta_t) = j | \alpha(t) = k, \alpha(t'), 0 \leq t' \leq t] = q_{jk}\Delta_t + o(\Delta_t), \quad \forall i \neq j. \tag{10}$$

Given the SV model in equation (5), and with variance process

$$\begin{cases} dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2, \\ v(0) = v_0, \end{cases} \tag{11}$$

we first model v_t as a m_0-state Markov chain $v_{\alpha(t)}$ with variance states $\mathbf{v} = \{v_1, v_2, \ldots, v_{m_0}\}$ satisfying $v_{j-1} < v_j$, indexed by $\mathcal{M} = \{1, \ldots, m_0\}$, as well as the corresponding generator Q governing the transition between variance states.

Once v_1 and v_{m_0} have been determined (see Section A.0.1 for details on choosing v_1 and v_{m_0}), we first define a non-uniform grid as in Tavella and Randall (2000):

$$v_j = v_0 + \hat{\alpha} \sinh\left(c_2 \frac{j}{m_0} + c_1\left(1 - \frac{j}{m_0}\right)\right), \quad j = 2, \ldots, m_0 - 1, \tag{12}$$

where

$$c_1 = \sinh^{-1}\left(\frac{v_1 - v_0}{\hat{\alpha}}\right), \quad c_2 = \sinh^{-1}\left(\frac{v_{m_0} - v_0}{\hat{\alpha}}\right) \tag{13}$$

for $\hat{\alpha} < (v_{m_0} - v_1)$; here $\hat{\alpha}$ determines the uniformity of the grid.[3] Let $\mathbf{k} = \{k_1, k_2, \ldots, k_{m_0-1}\}$ be the resulting set of grid spacings, with $k_i = v_i - v_{i-1}$.

Next we parameterize the generator Q as in Lo and Skindilias (2014) by

$$q_{ij} = \begin{cases} \dfrac{\hat{\mu}^-(v_i)}{k_{i-1}} + \dfrac{\hat{\sigma}^2(v_i) - (k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i))}{k_{i-1}(k_{i-1}+k_i)}, & \text{if } j = i-1, \\ \dfrac{\hat{\mu}^+(v_i)}{k_i} + \dfrac{\hat{\sigma}^2(v_i) - (k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i))}{k_i(k_{i-1}+k_i)}, & \text{if } j = i+1, \\ -q_{i,i-1} - q_{i,i+1}, & \text{if } j = i, \\ 0, & \text{if } |j-i| > 1, \end{cases} \tag{14}$$

[3]The use of a non-uniform grid is common in the partial differential equation (PDE) literature, and we find that it has more hastened convergence over a uniform grid. Alternative approaches may also be used instead.

where $a^{\pm} = \max\{\pm a, 0\}$. As long as $\mathbf{k}$ is chosen such that

$$0 < \max_{1 \leq i \leq m_0 - 1} k_i \leq \min_{1 \leq i \leq m_0} \frac{\hat{\sigma}^2(v_i)}{|\hat{\mu}(v_i)|}, \tag{15}$$

then we have

$$\hat{\sigma}^2(v_i) \quad \geq \quad k_{i-1}\hat{\mu}^-(v_i) + k_i\hat{\mu}^+(v_i), \tag{16}$$

so the set of q_{ij} is well defined, and the first two instantaneous moments of v_t are locally matched with those of the Markov chain:

$$\mathbb{E}[v_{t+\Delta_t} - v_t] = \hat{\mu}(v_i)\Delta_t, \quad \mathbb{E}[v_{t+\Delta_t} - v_t]^2 = (\hat{\sigma}(v_i))^2\Delta_t. \tag{17}$$

Hence the CTMC $v_{\alpha(t)}$ satisfies the local consistency conditions (see Kushner (1990)) and converges weakly to v_t.

3.2 Regime-Switching Model for the Underlying

Given a CTMC approximation of v_t, we represent the underlying model as a regime-switching jump diffusion, where regimes represent the variance states. Recall that the SV model in equation (5) can be written in terms of the reformulated process $\widetilde{X}_t = \log(S_t/S_0) - f(v_t, v_0)$, with dynamics

$$\begin{cases} d\widetilde{X}_t = \left(\gamma - \frac{1}{2}m^2(v_t) - \rho h(v_t)\right)dt + \sqrt{1-\rho^2}m(v_t)dW_t^* + d\left[\sum_{k=1}^{N_t} Y_k\right], \\ dv_t = \hat{\mu}(v_t)dt + \hat{\sigma}(v_t)dW_t^2, \end{cases} \tag{18}$$

where $\gamma = r - q - \lambda\kappa$. Note that $\kappa = \phi(-\mathrm{i}) - 1$, where $\phi(\xi)$ is the characteristic function of the jump size distribution. More generally, we can model the jump component as a pure jump Lévy process with a known characteristic function.

With variance states $v_t \in \{v_1, \ldots, v_{m_0}\}$ indexed by $j \in \mathcal{M} = \{1, \ldots, m_0\}$, the dynamics are approximated by the regime-switching process $\widetilde{X}_t^{m_0}$ satisfying

$$d\widetilde{X}_t^{m_0} = \mu_{\alpha(t)}dt + \sigma_{\alpha(t)}dW_t^* + d\left[\sum_{k=1}^{N_t} Y_k\right], \quad \alpha(t) \in \{1, \ldots, m_0\}, \tag{19}$$

where

$$\begin{cases} \mu_{\alpha(t)} = \gamma - \frac{1}{2}m^2(v_{\alpha(t)}) - \rho h(v_{\alpha(t)}), \\ \sigma_{\alpha(t)} = \sqrt{1-\rho^2}m(v_{\alpha(t)}), \end{cases} \tag{20}$$

and here W_t^* is a standard Brownian motion independent of W_t^2 that drives the volatility process. The necessary transforms as well as the drift and volatility functions $\mu_{\alpha(t)}$ and $\sigma_{\alpha(t)}$ for equation (19) are provided in Table 3 for many popular SV processes, which we have augmented with jumps.

The log-return process $\widetilde{X}_t^{m_0}$ can be described by its characteristic function in each state $j \in \mathcal{M}$. Given a time increment of size $\Delta_t > 0$, and $\xi \in \mathbb{R}$, define

$$\widetilde{\phi}^j_{\widetilde{X}^{m_0}_{\Delta_t}}(\xi) := \mathbb{E}[e^{\mathrm{i}\xi \widetilde{X}^{m_0}_{\Delta_t}} | \alpha(0 \le s \le \Delta_t) = j] := \exp(\psi_j(\xi)\Delta_t), \quad j \in \mathcal{M} \tag{21}$$

where $\widetilde{\phi}^j_{\widetilde{X}^{m_0}_{\Delta_t}}(\xi)$ is the characteristic function (chf) of a process which spends the entirety of Δ_t in state j, and $\psi_j(\xi)$ is its *Lévy symbol.* The set of symbols describing the dynamics of $\widetilde{X}_t^{m_0}$ are

$$\psi_j(\xi) = \mathrm{i}\xi\mu_j - \frac{1}{2}\xi^2\sigma_j^2 + \lambda(\phi(\xi) - 1), \quad j = 1, \dots, m_0, \tag{22}$$

where $\lambda(\phi(\xi) - 1)$ is state-independent. Given the set $\{\psi_j(\xi)\}_{j=1}^{m_0}$, the characteristic function of $\widetilde{X}^{m_0}_{\Delta_t}$ is known in closed-form.

Lemma 1 *(Buffington and Elliot (2002)) For $\Delta_t > 0$, the characteristic function of $\widetilde{X}^{m_0}_{\Delta_t}$ is given by the following in matrix form*

$$\mathbb{E}[\exp(\mathrm{i}\widetilde{X}^{m_0}_{\Delta_t}\xi)|\alpha(0)] = \mathbf{1}' \cdot \exp\left(\Delta_t\left(Q' + diag(\psi_1(\xi), \dots, \psi_{m_0}(\xi))\right)\right)\mathbb{I}(0), \tag{23}$$

where $\mathbf{1} \in \mathbb{R}^{m_0}$ is a unit (column) vector, and $\mathbb{I}(0) \in \mathbb{R}^{m_0}$ is a vector of zeros, except for the value 1 in the position $\alpha(0)$.

Remark 1 *While our main focus in this paper is a pricing methodology for SV models, the framework can also be applied to pure regime-switching models as well. In particular, let $B(t)$ be a standard Brownian motion independent of the CTMC $\alpha(t)$. In an m_0 state regime-switching model, the underlying price $S(t) = S^{m_0}(t)$ evolves under the risk-neutral measure as*

$$\frac{dS(t)}{S(t^-)} = (r - q - \lambda_{\alpha(t)}\kappa_{\alpha(t)})dt + \sigma_{\alpha(t)}dB(t) + \int_{\mathbb{R}}(e^y - 1)\pi^\alpha(dy, dt), \tag{24}$$

where π^α is the Poisson random measure, ν_α is the corresponding Lévy measure with jump intensity $\lambda(\alpha(t), dy) = \lambda_{\alpha(t)}\nu_{\alpha(t)}(y)dy$, and $\kappa_{\alpha(t)} = \int_{\mathbb{R}}(e^y - 1)\nu_{\alpha(t)}(y)dy$ is finite for each regime. The log return process $X_t = \log(S_t/S_0), t \in [0, T]$ satisfies

$$X_t = \int_0^t \zeta_{\alpha(t)}dt + \int_0^t \sigma_{\alpha(t)}dB(t) + \int_0^t\int_{\mathbb{R}} y\pi^\alpha(dy, dt), \tag{25}$$

where $\zeta_{\alpha(t)} = r - q - \frac{1}{2}\sigma^2_{\alpha(t)} - \lambda_{\alpha(t)}\kappa_{\alpha(t)}$ is the risk-neutral (convexity-adjusted) drift. Given a prespecified generator for $\alpha(t)$, the same methodology can be used while bypassing the variance approximation step.

Remark 2 *With $X^{m_0}_{\Delta_t} = \ln(S^{m_0}_{t+\Delta_t}/S^{m_0}_t)$ having the same distribution as $\widetilde{X}^{m_0}_{\Delta_t} + f(v_{\alpha(\Delta_t)}, v_{\alpha(0)})$, it follows that*

$$\begin{aligned}\mathbb{E}[e^{\mathrm{i}\xi X^{m_0}_{\Delta_t}}|\alpha(0) = j, \alpha(\Delta_t) = k] &= \mathbb{E}[e^{\mathrm{i}\xi \widetilde{X}^{m_0}_{\Delta_t}}|\alpha(0) = j, \alpha(\Delta_t) = k] \cdot \exp(\mathrm{i}\xi f(v_k, v_j)) \\ &= \mathcal{E}_{k,j}(\xi) \cdot \exp(\mathrm{i}\xi f(v_k, v_j)),\end{aligned}$$

Hull-White	$f(v_k, v_j)$	$\frac{2\rho}{\sigma_v}(\sqrt{v_k} - \sqrt{v_j})$
	μ_j	$(\frac{\rho\sigma_v}{4} - \frac{a_v\rho}{\sigma_v})\sqrt{v_j} - \frac{1}{2}v_j + (r - q - \lambda\kappa)$
	σ_j	$\sqrt{(1-\rho^2)v_j}$
Scott	$f(v_k, v_j)$	$\frac{\rho}{\sigma_v}(e^{v_k} - e^{v_j})$
	μ_j	$\rho(\frac{\eta}{\sigma_v}(v_j - \theta) - \frac{\sigma_v}{2})e^{v_j} - \frac{e^{2v_j}}{2} + (r - q - \lambda\kappa)$
	σ_j	$e^{v_j}\sqrt{(1-\rho^2)}$
Stein-Stein	$f(v_k, v_j)$	$\frac{1}{2}\frac{\rho}{\sigma_v}(v_k^2 - v_j^2)$
	μ_j	$(\frac{\rho\eta}{\sigma_v} - \frac{1}{2})v_j^2 - \frac{\rho\eta\theta}{\sigma_v}v_j + (r - q - \lambda\kappa - \frac{\rho\sigma_v}{2})$
	σ_j	$v_j\sqrt{(1-\rho^2)}$
Heston	$f(v_k, v_j)$	$\frac{\rho}{\sigma_v}(v_k - v_j)$
	μ_j	$(\frac{\rho\eta}{\sigma_v} - \frac{1}{2})v_j + (r - q - \lambda\kappa - \frac{\rho\eta\theta}{\sigma_v})$
	σ_j	$\sqrt{(1-\rho^2)v_j}$
3/2	$f(v_k, v_j)$	$\frac{\rho}{\widehat{\sigma}_v}\left(\log(\widehat{v}_k) - \log(\widehat{v}_j)\right)$
	μ_j	$\frac{1}{2}(\rho\widehat{\sigma}_v - 2\frac{\rho\widehat{\eta}\widehat{\theta}}{\widehat{\sigma}_v} - 1)\frac{1}{\widehat{v}_j} + (\frac{\rho\widehat{\eta}}{\widehat{\sigma}_v} + r - q - \lambda\kappa)$
	σ_j	$\sqrt{\frac{(1-\rho^2)}{\widehat{v}_j}}$
4/2	$f(v_k, v_j)$	$\frac{\rho}{\sigma_v}\big(a_S(v_k - v_j) + b_S(\log(v_k) - \log(v_j))\big)$
	μ_j	$(\frac{a_S\rho\eta}{\sigma_v} - \frac{a_S^2}{2})v_j + (\frac{\rho b_S\sigma_v - b_S^2}{2} - \frac{b_S\rho\eta\theta}{\sigma_v})\frac{1}{v_j} + (\frac{\rho\eta}{\sigma_v}(b_S - a_S\theta) + r - q - \lambda\kappa - a_S b_S)$
	σ_j	$(a_S\sqrt{v_j} + \frac{b_S}{\sqrt{v_j}})\sqrt{(1-\rho^2)}$
α-H	$f(v_k, v_j)$	$\frac{\rho}{\sigma_v}(e^{v_k} - e^{v_j})$
	μ_j	$\frac{\rho\theta}{\sigma_v}e^{(1+a_v)v_j} - \rho(\frac{\eta}{\sigma_v} + \frac{\sigma_v}{2})e^{v_j} - \frac{e^{2v_j}}{2} + (r - q - \lambda\kappa)$
	σ_j	$e^{v_j}\sqrt{(1-\rho^2)}$

Table 3: Transform for pricing with common SV models with jumps.

where $\mathcal{E}(\xi) = [\mathcal{E}_{k,j}(\xi)]$ *is the matrix exponential defined as* $\mathcal{E}(\xi) = \exp\left(\Delta_t\left(Q' + diag(\psi_1(\xi), \ldots, \psi_{m_0}(\xi)\right)\right)$. *To handle applications for pure regime-switching models (as in Remark 1) as well as SV, we introduce the notation*

$$\widetilde{\mathcal{E}}_{k,j}(\xi) := \begin{cases} \mathcal{E}_{k,j}(\xi), & \text{for regime-switching model,} \\ \mathcal{E}_{k,j}(\xi) \cdot \exp(\mathrm{i}\xi f(v_k, v_j)), & \text{for stochastic volatility model.} \end{cases} \tag{26}$$

Remark 3 *An important property of this weak approximation approach is that the characteristic function of the underlying SV model is not even required to employ a transform-based approach, thus our method can be applied to a broader class of SV models where the characteristic functions either do not exist, or have not been determined in closed-form in the literature. This distinguishes our approach from the traditional transform-based analysis in the literature, e.g. Duffie et al. (2000). Our approach does not need the characteristic function or the transition density of the variance process or the integrated variance process, as required in Zheng and Kwok (2014) and Lian et al. (2014).*

3.3 Weak convergence

Proposition 1 *If we choose* $Q = (q_{ij})$ *as in* (14), *then* $(v_{\alpha(t)}, \widetilde{X}_t^{m_0})$ *converges weakly to* $(v_t, \widetilde{X}_t)$.

Proof: First let $\bar{X}_t$ be the diffusion process whose dynamics is the same as $\widetilde{X}_t$ but without the jump component, and similarly for $\bar{X}_t^{m_0}$. We first prove that $(v_{\alpha(t)}, \bar{X}_t^{m_0})$ converges weakly to $(v_t, \bar{X}_t)$. To this end, let $\theta(v) = \gamma - \frac{1}{2}[m(v)]^2 - \rho h(v)$, and assume that $\bar{X}_t = x, v_t = v$. Since W_t^* and $W_t^{(2)}$ are independent Brownian motions, we have the infinitesimal generator of $(v_t, \bar{X}_t)$ given by

$$\mathcal{L}V(x,v) = \frac{1}{2}(1-\rho^2)[m(v)]^2\frac{\partial^2 V}{\partial x^2} + \theta(v)\frac{\partial V}{\partial x} + \frac{1}{2}\hat{\sigma}^2(v)\frac{\partial^2 V}{\partial v^2} + \hat{\mu}(v)\frac{\partial V}{\partial v} \tag{27}$$

Without loss of generality, we assume that we have a uniform grid $\{v_1, v_2, \ldots, v_{m_0}\}$ on[4] $[v_1, v_{m_0}]$ for v (details on the choices of v_1, v_{m_0} will be discussed in Section A.0.1), that is $k_i = v_i - v_{i-1} \equiv \Delta k$. With the constant step size, the generator $Q = (q_{ij})$ in (14) is reduced to

$$q_{i,i-1} = \frac{-\hat{\mu}(v_i)}{2(\Delta k)} + \frac{\hat{\sigma}^2(v_i)}{2(\Delta k)^2}, \quad q_{i,i} = -\frac{\hat{\sigma}^2(v_i)}{(\Delta k)^2}, \quad q_{i,i+1} = \frac{\hat{\mu}(v_i)}{2(\Delta k)} + \frac{\hat{\sigma}^2(v_i)}{2(\Delta k)^2},$$

and $q_{i,j} = 0$ if $|j-i| > 1$. Let $V_i = V(x, i\Delta k)$, and similarly for the first and the second derivatives, then we can approximate $\frac{\partial^2 V_i}{\partial v^2}$ by $\frac{V_{i+1}-2V_i+V_{i-1}}{(\Delta k)^2}$, while $\frac{\partial V_i}{\partial v}$ is approximated by $\frac{V_{i+1}-V_i}{\Delta k}$ if $\mu(v_i) > 0$ and by $\frac{V_{i-1}-V_i}{\Delta k}$ if $\mu(v_i) < 0$. We have

$$\begin{aligned} &\sup_{v\in[v_1,v_{m_0}]}\left|\frac{1}{2}\hat{\sigma}^2(v_i)\frac{\partial^2 V_i}{\partial v^2} + \hat{\mu}(v_i)\frac{\partial V_i}{\partial v} - [q_{i,i-1}V_{i-1} + q_{i,i}V_i + q_{i,i+1}V_{i+1}]\right| \\ &= \sup_{v\in[v_1,v_{m_0}]}\left|\frac{\hat{\sigma}^2(v_i)}{2(\Delta k)^2}\left[V_{i+1} + V_{i-1} - 2V_i - \frac{\partial^2 V_i}{\partial v^2}(\Delta k)^2\right] + \frac{\hat{\mu}^+(v_i)}{\Delta k}\left[V_{i+1} - V_i - \frac{\partial V_i}{\partial v}\Delta k\right]\right. \\ &\left. -\frac{\hat{\mu}^-(v_i)}{\Delta k}\left[V_{i-1} - V_i - \frac{\partial V_i}{\partial v}\Delta k\right]\right| \to 0 \quad \text{as} \quad \Delta k \to 0. \end{aligned} \tag{28}$$

The infinitesimal generator $\mathcal{L}V_i(x)$ of $\bar{X}_t^{m_0}$ with $\alpha(t) = i, \bar{X}_t^{m_0} = x$ given by

$$\mathcal{L}V_i(x) = \frac{1}{2}(1-\rho^2)[m(v_i)]^2\frac{\partial^2 V_i}{\partial x^2} + \theta(v_i)\frac{\partial V_i}{\partial x} + \sum_{j=1}^{m_0} q_{ij}V_j. \tag{29}$$

From (28), it is clear that $\mathcal{L}V_i(x) \to \mathcal{L}V(x,v)$ as $\Delta k \to 0$. Hence by Theorem 5.1 of Mijatović and Pistorius (2013) $(v_{\alpha(t)}, \bar{X}_t^{m_0})$ converges weakly to $(v_t, \bar{X}_t)$. Finally, because of the independence of the jump component we have

$$\begin{aligned} \mathbb{E}[\exp(\mathrm{i}\widetilde{X}_t^{m_0}\xi + \mathrm{i}v_{\alpha(t)}\eta)] &= \exp(t\lambda(\phi(\xi)-1))\mathbb{E}[\exp(\mathrm{i}\bar{X}_t^{m_0}\xi + \mathrm{i}v_{\alpha(t)}\eta)] \\ &\to \exp(t\lambda(\phi(\xi)-1))\mathbb{E}[\exp(\mathrm{i}\bar{X}_t\xi + \mathrm{i}v_t\eta)] \\ &= \mathbb{E}[\exp(\mathrm{i}\widetilde{X}_t\xi + \mathrm{i}v_t\eta)]. \end{aligned}$$

Hence, $(v_{\alpha(t)}, \widetilde{X}_t^{m_0})$ converges weakly to $(v_t, \widetilde{X}_t)$.[5] □

[4] As in Mijatović and Pistorius (2013), we assume that the grid bounds v_1, v_{m_0} are chosen so that $[v_1, v_{m_0}]$ will cover the domain of v_t as m_0 increases.

[5] The weak convergence on the level of the process implies in particular that the marginal distributions of $(v_{\alpha(t)}, \widetilde{X}_t^{(m_0)})$ will converge to those of $(v_t, \widetilde{X}_t)$, and therefore the values of the derivatives.

3.4 The $\tau/2$ Models

A key advantage of the proposed methodology is its generality. Consider for example the following variance dynamics

$$dv_t = \eta v_t^{\omega}(\theta - v_t)dt + \sigma_v v_t^{\tau/2} dW_t^2, \quad v(0) = v_0, \tag{30}$$

with $m(v_t) = \sqrt{v_t}$ in equation (5). With $\tau = 1$ and $\omega = 0$ we obtain Heston's model, while for $\tau = 3$ and $\omega = 1$ we obtain the 3/2 model. Allowing for arbitrary τ enables us to fit the model to empirical data (see for example Christoffersen et al. (2010)), without the usual restriction of analytical tractability for subsequent pricing. Ignoring the arbitrary constant, the transform for this class when $\tau \neq 3$ is simply

$$\widehat{f}(v) = \frac{2}{\sigma_v(3-\tau)} v^{(3-\tau)/2},$$

and

$$\begin{aligned} h(v) &= \eta v^{\omega}(\theta - v)\widehat{f}'(v) + \tfrac{\sigma_v^2}{2} v^{\tau} \widehat{f}''(v) \\ &= \frac{\eta}{\sigma_v} v^{(1-\tau)/2+\omega}(\theta - v) + \frac{\sigma_v}{4}(1-\tau)v^{(\tau-1)/2}. \end{aligned}$$

Equation (20) is then used to determine $\mu_{\alpha(t)}$ and $\sigma_{\alpha(t)} = \sqrt{(1-\rho^2)v_{\alpha(t)}}$. When $\omega = 0$ and $\tau = 2$, we obtain the "VAR" model, $dv_t = \eta(\theta - v_t)dt + \sigma_v v_t dW_t^2$, which contains Hull-White as a special case when $\theta = 0$ and $a_v = -\eta$. The VAR model is shown in Christoffersen et al. (2010) to outperform Heston's model for S&P500 data. They also provide empirical support for the 3/2 model ($\tau = 3$), for which we prefer an alternative form that results from a change of variables

$$\widehat{v}_t = \frac{1}{v_t}, \quad \widehat{\eta} = \eta\theta, \quad \widehat{\theta} = \frac{\eta + \sigma_v^2}{\eta\theta}, \quad \widehat{\sigma}_v = -\sigma_v. \tag{31}$$

In particular, we use the formulation

$$\begin{cases} dS_t = S_t(r - \lambda\kappa)dt + \dfrac{S_t}{\sqrt{\widehat{v}_t}} dW_t^1 + S_t(e^{J_t} - 1)dN_t, \\ d\widehat{v}_t = \widehat{\eta}[\widehat{\theta} - \widehat{v}_t]dt + \widehat{\sigma}_v \sqrt{\widehat{v}_t} dW_t^2, \quad \widehat{v}_0 = 1/v_0, \end{cases} \tag{32}$$

for which the mean and variance of $\widehat{v}_t$ are known and can be used to determine the variance grid bounds. Moreover, in this form the model is a special case of the 4/2 model formulation considered in Table 1.

4 Contracts on realized variance

4.1 General Approach

Consider the regime-switching framework introduced in Section 3.2, which applies equally well for pure regime-switching models. To ease notation, we write X_t in place of $X_t^{m_0}$, where the regime switching model is understood. We introduce a recursive method to calculate the characteristic function of a random sum which takes the form

$$f_M^w(\xi) := \mathbb{E}\left[\exp\left(\mathrm{i}\xi\sum_{m=1}^{M} w_m h(X_m)\right)\right] = \mathbb{E}\left[\exp\left(\mathrm{i}\xi Y_M^w\right)\right], \quad Y_M^w := \sum_{m=1}^{M} w_m h(X_m),$$

where $X_m := \ln(S_m/S_{m-1})$ is the log return, $h : \mathbb{R} \to \mathbb{R}$, and $w = \{w_m\}_{m=1}^M$ is a real-valued sequence of weights. Here X_m represents the log return process of a pure regime-switching model, or the Markov chain approximation of a SV model. We calculate the characteristic function recursively in terms of the sequences

$$Y_1 := w_M \cdot h(X_M), \quad Y_m := w_{M-(m-1)} \cdot h(X_{M-(m-1)}) + Y_{m-1}, \quad m = 2, \dots, M.$$

Conditional on $\alpha_{M-1} = j$, we initialize the recursion by

$$\begin{aligned}\phi_{Y_1}^j(\xi) = \mathbb{E}_{M-1}[e^{\mathrm{i}\xi Y_1}|\alpha_{M-1} = j] &= \mathbb{E}_{M-1}[e^{\mathrm{i}\xi w_M h(X_M)}|\alpha_{M-1} = j] \\ &= \mathbb{E}[e^{\mathrm{i}\xi w_M h(X_{\Delta_t})}|\alpha(0) = j],\end{aligned}$$

which we approximate by integrating against the density of X_{Δ_t} conditional on $\alpha(0)$, defined by its characteristic function from Lemma 1

$$\mathbb{E}[e^{\mathrm{i}\xi X_{\Delta_t}}|\alpha(0) = j] = \sum_{k=1,\dots,m_0} \widetilde{\mathcal{E}}_{k,j}(\xi).$$

In particular,

$$\phi_{Y_1}^j(\xi) = \int_{-\infty}^{\infty} \exp(i\xi w_M h(x)) p_j(x) dx,$$

where $p_j(x) = \mathbb{P}[X(\Delta_t) \in x + dx|\alpha(0) = j]$, which is independent of $X(0)$.

For $m = 2, \dots, M$, we let

$$P_{jk}^{\Delta t} = \mathbb{P}[\alpha(t + \Delta t) = k|\alpha(t) = j], \quad G_{jk} = \{\alpha(0) = j, \alpha(\Delta t) = k\},$$

and continue the recursion by

$$\begin{aligned}\phi_{Y_m}^j(\xi) &= \mathbb{E}_{M-m}[e^{\mathrm{i}\xi Y_m}|\alpha_{M-m} = j] \\ &= \mathbb{E}_{M-m}[\exp(\mathrm{i}\xi w_{M-m+1} h(X_{M-m+1})) \cdot e^{\mathrm{i}\xi Y_{m-1}}|\alpha_{M-m} = j] \\ &= \sum_{k=1,..,m_0} \mathbb{E}_{M-m}[\exp(\mathrm{i}\xi w_{M-m+1} h(X_{M-m+1})) \cdot e^{\mathrm{i}\xi Y_{m-1}}|\alpha_{M-m} = j, \alpha_{M-(m-1)} = k] \cdot P_{jk}^{\Delta t} \\ &= \sum_{k=1,..,m_0} \mathbb{E}_{M-m}[\exp(\mathrm{i}\xi w_{M-m+1} h(X_{M-m+1}))|G_{jk}] \\ &\qquad\qquad \cdot \mathbb{E}_{M-m}[e^{\mathrm{i}\xi Y_{m-1}}|\alpha_{M-m} = j, \alpha_{M-(m-1)} = k] \cdot P_{jk}^{\Delta t} \\ &= \sum_{k=1,..,m_0} \mathbb{E}[\exp(\mathrm{i}\xi w_{M-m+1} h(X_{\Delta_t}))|G_{jk}] \cdot \phi_{Y_{m-1}}^k(\xi) \cdot P_{jk}^{\Delta_t},\end{aligned}$$

where $\phi^k_{Y_{m-1}}(\xi) = \mathbb{E}_{M-m}[e^{\mathrm{i}\xi Y_{m-1}}|\alpha_{M-m} = j, \alpha_{M-(m-1)} = k]$ is the conditional characteristic function of Y_{m-1}, which is known at stage m. To obtain $\mathbb{E}[\exp(\mathrm{i}\xi w_{M-m+1}h(X_{\Delta_t}))|G_{jk}]$, for each $j, k = 1, \dots, m_0$, conditional on $\alpha(0)$ and $\alpha(\Delta_t)$, we will integrate the densities of X_{Δ_t}, which are defined by their characteristic functions

$$\mathbb{E}[e^{\mathrm{i}\xi X_{\Delta_t}}|G_{jk}] = \frac{1}{P^{\Delta_t}_{jk}}\widetilde{\mathcal{E}}_{kj}(\xi).$$

We denote the density of X_{Δ_t}, conditioned on $G_{j,k}$, by $p_{j,k}(y|x)$:

$$p_{j,k}(y|x) = \mathbb{P}[X_{\Delta_t} \in y + dy|X_0 = x, \alpha(0) = j, \alpha(\Delta_t) = k], \quad j, k = 1, \dots, m_0. \tag{33}$$

While $p_{j,k}(y|x) = p_{j,k}(y-x)$ does not have a closed-form expression, it can be obtained through $\mathbb{E}[e^{\mathrm{i}\xi X_{\Delta_t}}|G_{jk}]$ using Fourier methods. As a particular application, which has fast convergence for highly peaked densities as in the present case, we apply the PROJ method below.

4.1.1 Recursion for Realized Variance

As an immediate application with $w_m = 1$ and $h(x) = x^2$, we obtain the characteristic function of the discretely sampled variance

$$\phi_{Y_M}(\xi) := \mathbb{E}\left[\exp\left(\mathrm{i}\xi\sum_{m=1}^{M} X_m^2\right)\right] = \mathbb{E}\left[\exp\left(\mathrm{i}\xi Y_M\right)\right], \quad Y_M := \sum_{m=1}^{M} X_m^2, \tag{34}$$

where $V_M = \frac{1}{T}Y_M$ from (1). In this case, the recursion simplifies to

$$Y_1 := h(X_M), \quad Y_m := h(X_{M-(m-1)}) + Y_{m-1}, \quad m = 2, \dots, M. \tag{35}$$

Once $\phi^j_{Y_1}(\xi) = \mathbb{E}[e^{\mathrm{i}\xi h(X_{\Delta_t})}|\alpha(0) = j] = \sum_{k=1,..,m_0}\Phi_{j,k}(\xi)$ is determined,

$$\phi^j_{Y_m}(\xi) = \sum_{k=1,..,m_0}\Phi_{j,k}(\xi)\cdot\phi^k_{Y_{m-1}}(\xi), \quad m = 2, \dots, M, \tag{36}$$

where we define

$$\Phi_{j,k}(\xi) := P^{\Delta_t}_{j,k}\cdot\mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}))|G_{jk}], \quad j, k = 1, \dots, m_0. \tag{37}$$

The cost to recover $\phi^j_{Y_M}(\xi_n)$ for a ξ-grid of size N is therefore $\mathcal{O}(MNm_0^2)$ after $\Phi_{j,k}(\xi)$ are initialized. This method will apply as well to contracts on higher order (centered) moments with $h(x) = (x-c)^d$, allowing the fair strike determination and pricing of contracts on realized skewness and kurtosis.

Remark 4 *The discretely sampled variance in equation* (34) *is given in terms of squared log returns. One may also consider the relative returns*

$$\frac{S_m - S_{m-1}}{S_{m-1}} = \frac{S_m}{S_{m-1}} - 1 = \exp(X_m) - 1,$$

in which case the same algorithm applies with $h(x) := (\exp(x) - 1)^2$. This return structure appears in the payoff of "arithmetic variance swaps" (see Leontsinisa and Alexander (2016)), and our valuation methodology also applies to their case.

The following result characterizes the convergence of the ChF recursion for the regime switching approximation to stochastic volatility. The main assumption is that weak convergence can be described by an algebraic convergence rate for characteristic functions (e.g. equation (37)), which is empirically observed as illustrated in section 5.1.

Proposition 2 *Let M be a fixed number of monitoring dates, and $v(0) = v_j$ denote the initial variance state. With $Y_M := \sum_{m=1}^{M} h(X_m)$ (e.g. $h(x) = x^2$), let*

$$\hat{\phi}^j_{Y_M}(\xi) = \mathbb{E}\left[\exp\left(\mathrm{i}\xi Y_M\right)|v(0) = v_j\right]$$

denote the true ChF under the SV dynamics in (5). Fix a grid $\{v_j\}_{j=1}^{m_0}$ of variance states as in section A.0.1 with gridwidth parameter $\gamma > 0$, and recall that $\phi^j_{Y_M}(\xi)$ is the regime-switching approximation. Then

$$\max_{\xi_n} |\hat{\phi}^j_{Y_M}(\xi_n) - \phi^j_{Y_M}(\xi_n)| = \mathcal{O}(d(\gamma) + m_0^{-d}),$$

where $d(\gamma)$ is the truncation error in variance space and $d > 0$ is the dominant algebraic order of weak convergence, defined in the proof below.[6] *With truncation error controlled, the approximation decays as $\mathcal{O}(m_0^{-d})$.*

Proof: Denote the CTMC approximations with m_0 states by $v^{m_0}_{\alpha(t)}$ and $X^{m_0}_{\Delta_t} = \widetilde{X}^{m_0}_{\Delta_t} + f(v^{m_0}_{\alpha(\Delta_t)}, v^{m_0}_{\alpha(0)})$, where $\Delta_t = T/M$, and let $(v(t), X_t)$ denote the true SV process.[7] Denote the true ChF of the SV process by $\hat{\phi}^k_m(\xi) = \mathbb{E}[Y_m|v(t_{M-(m-1)}) \in n_k]$, when the sequence $\{Y_m\}$ from (35) is constructed in terms of the true SV model. Let $\phi^j_m(\xi) \equiv \phi^j_{Y_m}(\xi)$ be the set of ChFs corresponding to the regime switching approximation. In the first stage ($m = 1$), $\hat{\phi}^j_1(\xi) = \mathbb{E}[\exp(\mathrm{i}\xi h(X_{t_M} - X_{t_{M-1}}))|v(t_{M-1}) = v_j] = \mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}))|v(0) = v_j]$. To account for truncation error in the variance space, we first extend the CTMC grid $\{v_j\}_{j=1}^{m_0} \subset \{v_j\}_{n\in\mathbb{N}}$, and let $\{n_j\}_{n\in\mathbb{N}}$ denote a corresponding partition of the state space, where n_j is a neighborhood around v_j (i.e. $v_j \in n_j$) which does not overlap with its neighbors.[8] We then have

$$\begin{aligned} P^{\Delta_t}_{j,k} = \mathbb{Q}[v^{m_0}_{\alpha(\Delta_t)} = v_k|v^{m_0}_{\alpha(0)} = v_j] &= \mathbb{Q}[v^{m_0}_{\alpha(\Delta_t)} \in n_k|v^{m_0}_{\alpha(0)} \in n_j] \\ &\approx \mathbb{Q}[v(\Delta_t) \in n_k|v(0) \in n_j] := \hat{P}^{\Delta_t}_{j,k}, \end{aligned} \tag{38}$$

[6] We assume that each source of error caused by the regime switching approximation decays at some algebraic rate. Here, $d > 0$ is the slowest such rate of convergence.

[7] Note from the continuous mapping theorem, $X^{m_0}_{\Delta_t} = \widetilde{X}^{m_0}_{\Delta_t} + f(v^{m_0}_{\alpha(\Delta_t)}, v^{m_0}_{\alpha(0)}) \Longrightarrow \widetilde{X}_{\Delta_t} + f(v(\Delta_t), v(0)) = X_{\Delta_t}$.

[8] For example, $n_j = (v_j - \epsilon/2, v_j + \epsilon/2]$ on a uniform grid with grid spacing $\epsilon = v_j - v_{j-1}$.

where $\hat{P}^{\Delta_t}_{j,k}$ is the true probability function, which follows from weak convergence.[9]

Next define $\hat{\Phi}_{j,k}(\xi) = \hat{P}^{\Delta_t}_{j,k} \cdot \mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}))|\hat{G}_{jk}]$ as the true ChF with $\hat{G}_{jk} = \{v(0) \in n_j, v(\Delta_t) \in n_k\}$. With error term $\hat{\mathcal{E}}^j_m(\xi) = \hat{\phi}^j_m(\xi) - \phi^j_m(\xi)$, we have

$$\begin{aligned}\hat{\mathcal{E}}^j_m(\xi) &= \sum_{k\in\mathbb{N}} \hat{\Phi}_{j,k}(\xi)\hat{\phi}^k_{m-1}(\xi) - \sum_{k=1,..,m_0} \Phi_{j,k}(\xi)\phi^k_{m-1}(\xi) \\ &= d^j_m(\xi) + \sum_{k=1,..,m_0} \left(\hat{\Phi}_{j,k}(\xi)\hat{\phi}^k_{m-1}(\xi) - \Phi_{j,k}(\xi)\phi^k_{m-1}(\xi)\right)\end{aligned}$$

where $d^j_m(\xi)$ represents truncation error for the series. Now define $e_{j,k}$ such that $P^{\Delta_t}_{j,k} = \hat{P}^{\Delta_t}_{j,k}(1+e_{j,k})$, where we assume from weak convergence that $\exists d_1 > 0$ such that $\max_{\{j,k\}} e_{j,k} = \mathcal{O}(m_0^{-d_1})$. With $E_{j,k}(\xi) := \mathbb{E}[\exp(\mathrm{i}\xi h(X^{m_0}_{\Delta_t}))|G_{jk}]$, and $\hat{E}_{j,k}(\xi) := \mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}))|\hat{G}_{jk}]$ it holds

$$\begin{aligned}\Phi_{j,k}(\xi) - \hat{\Phi}_{j,k}(\xi) &= P^{\Delta_t}_{j,k} \cdot E_{j,k}(\xi) - \hat{P}^{\Delta_t}_{j,k} \cdot \hat{E}_{j,k}(\xi) \\ &= \hat{P}^{\Delta_t}_{j,k}(1+e_{j,k}) \cdot E_{j,k}(\xi) - \hat{P}^{\Delta_t}_{j,k} \cdot \hat{E}_{j,k}(\xi) \\ &= \hat{P}^{\Delta_t}_{j,k}\left(e_{j,k} \cdot E_{j,k}(\xi) + (E_{j,k}(\xi) - \hat{E}_{j,k}(\xi))\right).\end{aligned}$$

From $G_{j,k} := \{v^{m_0}_{\alpha(0)} = v_j, v^{m_0}_{\alpha(\Delta_t)} = v_k\} = \{v^{m_0}_{\alpha(0)} \in n_j, v^{m_0}_{\alpha(\Delta_t)} \in v_k\}$, we have

$$\begin{aligned}E_{j,k}(\xi) &= \mathbb{E}[\exp(\mathrm{i}\xi h(X^{m_0}_{\Delta_t}))|\{v^{m_0}_{\alpha(0)} \in n_j, v^{m_0}_{\alpha(\Delta_t)} \in n_k\}] \\ &\approx \mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}))|\{v(0) \in n_j, v(\Delta_t) \in n_k\}] = \hat{E}_{j,k}(\xi)\end{aligned}$$

and from weak convergence[10]$\max_{\xi_n,\{j,k\}} |E_{j,k}(\xi_n) - \hat{E}_{j,k}(\xi_n)| = \mathcal{O}(m_0^{-d_2})$, for some $d_2 > 0$. Similarly, with $|E_{j,k}(\xi)| \leq 1$ we have $e_{j,k} \cdot E_{j,k}(\xi) = \mathcal{O}(m_0^{-d_1})$. Hence $|\Phi_{j,k}(\xi) - \hat{\Phi}_{j,k}(\xi)| = \hat{P}^{\Delta_t}_{j,k} \cdot \mathcal{O}(m_0^{-d_2} + m_0^{-d_1}) \leq \hat{P}^{\Delta_t}_{j,k} \cdot A_2 m_0^{-d}$, for some constant A_2 and $d := \min\{d_1, d_2\}$. We thus decompose

$$|\hat{\mathcal{E}}^j_m(\xi)| \leq |d^j_m(\xi)| + \sum_{k=1,..,m_0} \hat{P}^{\Delta_t}_{j,k}|\hat{\phi}^k_{m-1}(\xi) - \phi^k_{m-1}(\xi)| + \sum_{k=1,..,m_0} |\phi^k_{m-1}(\xi)||\hat{\Phi}_{j,k}(\xi) - \Phi_{j,k}(\xi)|,$$

and with $|\phi^k_{m-1}(\xi)| \leq \zeta_2(\xi) := \max_{1\leq m\leq M}\{\max_{1\leq k\leq m_0} |\phi^k_m(\xi)|\} \leq 1$ we have

$$\max_{1\leq j\leq m_0} |\hat{\mathcal{E}}^j_m(\xi)| \leq d(\gamma) + \zeta_2(\xi)\hat{\mathcal{E}}^*_{m-1} + A_2 m_0^{-d}$$

where $d(\gamma)$ is the maximum truncation error in variance space (which depends on the variance grid parameter γ), and $\hat{\mathcal{E}}^*_m = \max_{1\leq j\leq m_0}\{\max_{\xi_n}\{|\hat{\mathcal{E}}^j_m(\xi_n)|\}\}$. Hence

$$\max_{1\leq j\leq m_0} |\hat{\mathcal{E}}^j_M(\xi_n)| \leq \hat{\mathcal{E}}^*_1 \cdot \sum_{m=0}^{M-1} \zeta_2(\xi_n)^m \tag{39}$$

where $\hat{\mathcal{E}}^*_1 \leq d(\gamma) + A_2 m_0^{-d}$ (note that $0 \leq \zeta_2(\xi) << 1$ for large ξ). Also note that $\hat{\mathcal{E}}^j_M(\xi_1) = 0$ by design (i.e. the ChF at zero is one), and $\zeta_2(\xi_n) < 1$ for $n \geq 2$. This completes the proof. □

[9] Weak convergence (for the conditional measure) is equivalent to $\mu_{m_0}(A|n_j) \to \mu(A|n_j)$ for sets of the form $A = (-\infty, b]$ where $\mu(\{b\}) = 0$ (Billingsley (2008)). For m_0 fixed, $\mu_{m_0}(A|n_j)$ denotes the measure of $v^{m_0}_{\alpha(\Delta_t)}|\{v^{m_0}_{\alpha(0)} \in n_j\}$, and $\mu(A|n_j)$ is the true measure of $v(\Delta_t)|\{v(0) \in n_j\}$. Note too that $\mu_{m_0}(n_k|n_j) = P^{\Delta_t}_{j,k}$, since in discrete the CTMC world, $n_k = v_k$.

[10] Note that for continuous $h(x)$, $(h(X^{m_0}_t), v_{\alpha(t)}) \Longrightarrow (h(X_t), v_t)$ by the continuous mapping theorem.

4.1.2 Characteristic function Approximation

While the described methodology is independent of the numerical implementation, we will adopt the method of biorthogonal (frame) projection introduced in Kirkby (2015) to approximate the required transition densities. This method, called PROJ (for projection), represents the probability density of a random variable as the orthogonal projection onto a suitably chosen biorthogonal basis (see Section A.0.2 for an overview). Since then it has been extended to American and barrier options under very general dynamics (see Kirkby (2017a); Kirkby et al. (2017)).

The characteristic function recursion in Section 4.1.1 defines a sequence of ChFs, ending with $\phi^j_{Y_M}(\xi)$. We return to the initialization of $\phi^j_{Y_1}(\xi)$, but first suppose that $2 \leq m \leq M$. The main calculation in the ChF $\phi^j_{Y_m}(\xi)$ is to find $\Phi_{j,k}$ in equation (37). We first approximate the conditional transition density in equation (33) by its orthogonal projection onto the cubic basis

$$\begin{aligned} p_{j,k}(\nu) &\approx \sum_{n\in\mathbb{Z}} \left(\int_{-\infty}^{\infty} p_{j,k}(\eta) \widetilde{\varphi}_{a,n}(\eta) d\eta \right) \varphi_{a,n}(\nu) \\ &= \frac{a^{-1/2}}{\pi} \frac{1}{P^{\Delta_t}_{j,k}} \sum_{n\in\mathbb{Z}} \Re \left\{ \int_0^{\infty} \exp(-i\nu_n \xi) \cdot \widetilde{\mathcal{E}}_{k,j}(\xi) \cdot \widehat{\widetilde{\varphi}}(\xi/a) d\xi \right\} \varphi_{a,n}(\nu). \end{aligned} \tag{40}$$

With the number of basis element N fixed (below we set $N = 2^{10}$), and a grid width parameter $\bar{\alpha} \in \mathbb{R}$ selected as in Remark 8, we define a common grid for the density projections by

$$\Delta = 2\bar{\alpha}/(N-1), \quad x_1 = (1 - N/2)\Delta, \quad x_n = x_1 + (n-1)\Delta, \quad n = 1, \dots, N,$$

and define $a = 1/\Delta$. The frequency grid $\{\xi_n\}_{n=1}^N$ is then determined by equation (52). The analytical formula in equation (40) is discretized by

$$P^{\Delta_t}_{j,k} \cdot p_{j,k}(\nu) \approx a^{1/2} \Upsilon_{a,N} \sum_{1\leq l\leq N} \bar{\beta}^{j,k}_{a,l} \cdot \varphi_{a,l}(\nu) =: \bar{p}_{j,k}(\nu),$$

where $\left\{ \bar{\beta}^{j,k}_{a,l} \right\}_{l=1}^N := \Re \left\{ \mathcal{D} \left\{ H^{j,k}_n \right\} \right\}$ is determined by the DFT input vector

$$H^{j,k}_1 := \widetilde{\mathcal{E}}_{k,j}(\xi_1)/32a^4, \qquad H^{j,k}_n := \widetilde{\mathcal{E}}_{k,j}(\xi_n) \cdot \zeta_n \cdot \exp(-\mathrm{i}\xi_n \cdot x_1), \quad n \geq 2 \tag{41}$$

with ζ_n defined in equation (55). Given a representation of the transition densities between each state, the conditional characteristic function of $h(X_{\Delta_t})$ can be determined along $\{\xi_n\}_{n=1}^N$.

For each ξ_n, we approximate $\Phi(\xi_n)$ by $\bar{\Phi}(\xi_n) = \{\bar{\Phi}_{j,k}(\xi_n)\}_{j,k=1}^{m_0}$ which is obtained by

$$
\begin{aligned}
\Phi_{j,k}(\xi_n) &= P_{j,k}^{\Delta_t} \cdot \mathbb{E}[\exp(\mathrm{i}\xi_n h(X_{\Delta_t}))|G_{jk}] \\
&= P_{j,k}^{\Delta_t} \int_{-\infty}^{\infty} e^{\mathrm{i}\xi_n h(x)} p_{j,k}(x)dx \\
&\approx \Upsilon_{a,N} \sum_{1\le l\le N} \bar{\beta}_{a,l}^{j,k} \cdot a^{1/2} \int_{-\infty}^{\infty} e^{\mathrm{i}\xi_n h(x)} \varphi_{a,l}(x)dx \\
&= \Upsilon_{a,N} \sum_{1\le l\le N} \bar{\beta}_{a,l}^{j,k} \Psi_{n,l} \\
&= \Psi_{(n,\cdot)} B^{j,k} =: \bar{\Phi}_{j,k}(\xi_n),
\end{aligned} \tag{42}
$$

where $B^{j,k}$ is a vector[11] with $B_l^{j,k} := \bar{\beta}_{a,l}^{j,k}$ and Ψ is the matrix of integrals

$$
\Psi_{n,l} = \Upsilon_{a,N} \cdot a^{1/2} \int_{-\infty}^{\infty} e^{\mathrm{i}\xi_n h(x)} \varphi_{a,l}(x)dx, \quad n,l = 1,\dots,N,
$$

which are stored at initialization. We return to the computation of Ψ below. In matrix notation $\bar{\Phi}_{j,k} = \Psi B^{j,k}$. After determining $\bar{\Phi}_{j,k}$, the sequence of $\phi_{Y_m}^j(\xi_n)$ for $m = 2,\dots,M$ satisfies equation (36). Returning to the initialization step, $\phi_{Y_1}^j(\xi)$ is similarly determined by

$$
\begin{aligned}
\phi_{Y_1}^j(\xi) &= \sum_{k=1,..,m_0} \mathbb{E}[\mathbb{1}_{\{\alpha(\Delta_t)=k\}} \cdot \exp(\mathrm{i}\xi h(X_{\Delta_t}))|\alpha(0) = j] \\
&= \sum_{k=1,..,m_0} \Phi_{j,k}(\xi),
\end{aligned} \tag{43}
$$

which is approximated by $\bar{\phi}_{Y_1}^j(\xi)$, by replacing Φ with $\bar{\Phi}$ in (43).

4.1.3 Final Step: Payoff Integration

In the final stage, the recursion terminates with the set $\{\phi_{Y_M}^j(\xi)\}_{j=1}^{m_0}$. Given the initial variance v_0, we find the bracketing variance states, indexed by k_0 and k_0+1 such that $v_{k_0} \le v_0 < v_{k_0} + 1$. With a terminal grid $\{y_n\}_{n=1}^{N/2}$ for Y_M fixed below, we recover the density coefficients[12] corresponding to $\phi_{Y_M}^j(\xi)$ for $j \in \{k_0, k_0+1\}$, which we denote by $\{\hat{\beta}_{a,n}^j\}_{n=1}^N$. They are found by using the FFT as in equation (53), where the input for $j \in \{k_0, k_0+1\}$ is

$$
H_1^j := 1/32a^4, \qquad H_n^j := \phi_{Y_M}^j(\xi_n) \cdot \zeta_n \cdot \exp(-\mathrm{i}\xi_n \cdot y_1), \quad n \ge 2, \tag{44}
$$

with ξ_n defined in equation (52) and ζ_n in (55).

The remaining calculation depends on the payoff (e.g. variance swap, call on variance or volatility). For a terminal function $f(\cdot)$ of Y_M, the final expectation is the interpolated value

$$
\mathbb{E}[f(Y_M)] \approx V^{k_0} + (V^{k_0+1} - V^{k_0}) \frac{v_0 - v_{k_0}}{v_{k_0+1} - v_{k_0}}
$$

[11]These vectors will not be stored, since they are needed only once.

[12]For simplicity, we use a cubic basis for all computations. However, given the near singular behavior of the left tail of Y_M, switching to the linear basis in the final projection stage can be used to improve performance.

where

$$V^j := \Upsilon_{a,N} \sum_{n=1}^{N/2} \hat{\beta}^j_{a,n} \theta^f_n, \quad j = k_0, k_0 + 1,$$

and

$$\theta^f_n := a^{1/2} \int_0^\infty \varphi_{a,n}(y) f(y) dy.$$

For a call option on realized variance with strike K, we have

$$C = \tfrac{e^{-rT}}{T} \mathbb{E}[(Y_M - KT)^+] = \tfrac{e^{-rT}}{T} \mathbb{E}[f(Y_M)].$$

First fix the terminal grid, defined by $y_1 = KT - \Delta$, $y_n = y_1 + (n-1)\Delta$ for $n = 1, \ldots, N/2$ (we use $N/2$ because of possible boundary effects, since the terminal density is restricted to $[0, \infty)$, and has sharp peaks near the origin). We have the set of coefficients corresponding to $f(y) = (y - KT)^+$ given by

$$\theta_n = \begin{cases} (y_n - KT)/24 + \Delta/20 & n = 1 \\ (y_n - KT)/2 + \Delta \cdot 7/30 & n = 2 \\ (y_n - KT) \cdot 23/24 + \Delta/20 & n = 3 \\ (y_n - KT) & n = 4, \ldots, N/2, \end{cases} \tag{45}$$

where we note that $y_n - KT = -\Delta + (n-1)\Delta$. The put option coefficients can be similarly derived.

The coefficients of a variance swap $K_{Var} = \mathbb{E}[V_M] = \frac{1}{T}\mathbb{E}[Y_M]$ are identical to those of call option with $K = 0$. Hence, the swap grid is determined by $y_1 = -\Delta$, and the coefficients in equation (45) are applied with $K = 0$. An advantage of our approach of ChF recovery is that once the set $\phi^j_{Y_M}(\xi)$ is recovered, we can obtain prices for an entire spectrum of prices at negligible additional cost (equivalent to the cost of an European option valuation with known ChF).

Note that in the final stage M of the algorithm, $\{v_m\}_{m=1}^{m_0}$ represents a discrete set of initial states for the variance process, from which the value for v_0 is obtained by interpolation. The left panel of Figure 1 illustrates the recovered realized variance density for a subset of the initial variance states in Heston's model, along with the modulus of the corresponding ChF, $|\phi^j_{Y_M}(\xi)|$, in the right panel. As initial variance v_m increases, the rate of decay of $|\phi^j_{Y_M}(\xi)|$ also increases, resulting in a smoother density of realized variance.

Remark 5 *We can also price volatility swaps and options on volatility, with grids similarly defined as for the variance counterparts. The difference lies in the calculation of θ_n. For example, the coefficients to determine the fair strike $K_{Vol} = \mathbb{E}[\sqrt{V_M}] = \frac{1}{\sqrt{T}}\mathbb{E}[\sqrt{Y_M}]$ of a volatility swap with $f(y) = \sqrt{y}$ are*

$$\theta^f_n := a^{1/2} \int_0^\infty \varphi_{a,n}(y) f(y) dy = \int_{[a_n, b_n]} \varphi(y) f(x_n + y/a) dy,$$

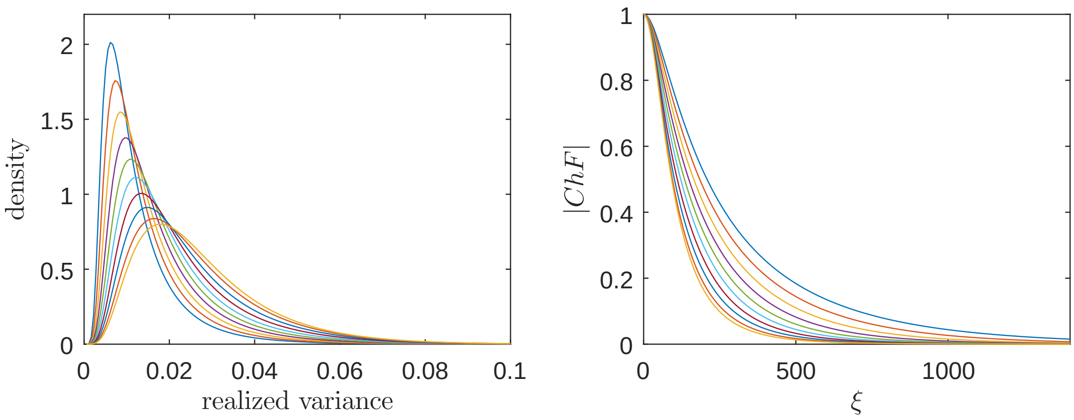

Figure 1: (Left) Densities of realized variance as a function of v_0. Initial variance ranges from $1e-05$ (most peaked density) to 0.0771 (smoothest density), where $m_0 = 40$. Heston model: $T = 1$, $m = 12$, $\eta = 3.99$, $\theta = 0.014$, $\rho = -0.79$, $\sigma_v = 0.27$, $r = 0.0319$. (Right) Modulus of corresponding ChFs, $|\phi^j_{Y_M}(\xi)|$ for $j = 4, 8, \ldots, m_0$.

where $[a_n, b_n] = [1, 2]$ *for* $n = 1$, $[0, 2]$ *for* $n = 2$, $[-1, 2]$ *for* $n = 3$, *and* $[-2, 2]$ *for* $n = 4, \ldots, N$. *Gaussian quadrature can then be applied analogously to the determination of* $\Psi(n, k)$, *with a five-point rule applied over each interval of unit length. For the call on volatility, we have*

$$C = e^{-rT}\mathbb{E}\left[(\sqrt{V_M} - \sqrt{K})^+\right] = \frac{e^{-rT}}{\sqrt{T}}\mathbb{E}\left[(\sqrt{Y_M} - \sqrt{KT})^+\right]$$

with $f(y) = (\sqrt{y} - \sqrt{KT})^+$, *and the grid can be fixed by* $y_1 = KT - \Delta$.

Remark 6 *In-progress options can also be easily priced in this framework. Let* T *denote the time remaining until maturity, and* V_0 *the current realized variance as of time zero. Considering for example the call on variance, we have*

$$C = e^{-rT}\mathbb{E}(V_0 + V_M - K)^+ = \frac{e^{-rT}}{T}\mathbb{E}(Y_M - (K - V_0)T)^+,$$

which is priced as a call with strike $(K - V_0)T$ *when* $K \geq V_0$. *Otherwise,* $C = \frac{e^{-rT}}{T}(\mathbb{E}[Y_M] + (V_0 - K)T)$, *for which we calculate* $\mathbb{E}[Y_M]$ *as a swap contract.*

The PROJ method for variance/volatility swaps and options is summarized in Algorithm 1, and it applies to RS and SV models. For pure RS models, m_0 and the generator Q are supplied rather than determined as for SV models. Moreover, for RS models the initial state k_0 is supplied and interpolation is not required. Note that in the loop $m = 2, \ldots, M$, we define the set of vectors $\boldsymbol{\phi}(\xi_n) = \{\phi^j(\xi_n)\}_{j=1}^{m_0}$, so equation (36) reduces to a set of matrix-vector multiplications. See Section 6.1 for a simple extension to Lévy models.

We now analyze the complexity of Algorithm 1. During initialization, there are N matrix exponential calculations at a cost of $\mathcal{O}(Nm_0^3)$ to populate $\widetilde{\mathcal{E}}$ and order $\mathcal{O}(N^2)$ calculations for $\bar{\Psi}$. For each $j, k = 1, \ldots, m_0$ fixed, $\bar{\Phi}_{j,k}$ requires the calculation of $\{B_l^{j,k}\}_{l=1}^N$ at a cost

Algorithm 1 PROJ Algorithm for RS and SV models

Fix N, L_1, m_0, γ
Determine $\bar{\alpha}$ as in Remark 8
For SV models: determine the grid $\{v_j\}_{j=1}^{m_0}$ and initialize generator Q (Section 3.1)
Compute ζ_n, $\quad n = 1, \dots, N \qquad$ (eq. (55))
Matrix Exponential: $\widetilde{\mathcal{E}}_{j,k}(\xi_n)$, $j, k = 1, \dots, m_0, \quad n = 1, \dots, N \quad$ (eq. (26))
Psi Matrix: $\bar{\Psi}(\xi_n, k)$, $n, k = 1, \dots, N, \quad$ (Section A.0.3)
Phi Matrix: $\bar{\Phi}_{j,k}(\xi_n)$, $j, k = 1, \dots, m_0, \quad n = 1, \dots, N \quad$ (Section 4.1.2)
Initial ChF: $\boldsymbol{\phi}(\xi_n) \leftarrow \bar{\Phi}(\xi_n)\mathbf{1}, \quad n = 1, \dots, N$ (Section 4.1.2)
for $m = 2, \dots, M$ **do**
  for $n = 1, \dots, N$ **do**
    $\boldsymbol{\phi}(\xi_n) \leftarrow \bar{\Phi}(\xi_n)\boldsymbol{\phi}(\xi_n) \quad$ (eq. (36))
  end for
end for
Final valuation (variance/volatility swaps and options), see Section 4.1.3

$\mathcal{O}(m_0^2 \cdot N \log_2(N))$, followed by the inner products $\bar{\Phi}(\xi_n) = \Psi_{(n,\cdot)} B^{j,k}$, at a total cost of $\mathcal{O}(m_0^2 \cdot N^2)$ for all $\{\xi_n\}_{n=1}^N$, which is the dominant cost of $\bar{\Phi}$. Hence, initialization requires $\mathcal{O}(m_0^2 \cdot N^2 + N \cdot m_0^3)$ operations. The main loop requires $\mathcal{O}(m_0^2 \cdot N \cdot M)$ operations, for a full algorithm complexity of $(m_0^2 \cdot N^2 + N \cdot m_0^3 + m_0^2 \cdot N \cdot M)$. With N, m_0 fixed, the incremental cost is linear in M.

5 Error Analysis

Proposition 3 *Let M be a fixed number of monitoring dates, and $v(0) = v_j$ denote the initial variance state. With $Y_M := \sum_{m=1}^M h(X_m)$, let $\hat{\phi}^j_{Y_M}(\xi)$ denote the true ChF under the SV dynamics in* (5)*, and recall that $\bar{\phi}^j_{Y_M}(\xi)$ is the regime-switching approximation in the presence of projection error. Then*

$$\max_{\xi_n} |\hat{\phi}^j_{Y_M}(\xi_n) - \bar{\phi}^j_{Y_M}(\xi_n)| = \mathcal{O}(d(\gamma) + m_0^{-d} + \bar{\alpha}^{9/2} N^{-4} + \tau(\bar{\alpha})),$$

where $\tau(\bar{\alpha})$ is the truncation error of the density projection, while $d > 0$ and $d(\gamma)$ are defined in Proposition 2. Hence, with truncation errors $d(\gamma)$ and $\tau(\bar{\alpha})$ controlled in the variance and log asset space (resp.), the error decays with increasing m_0 and N.

Proof: We now prove the recursive error convergence for the full approximation scheme, where we continue to assume the notation from the proof of Proposition 2. Let $\bar{\phi}^j_1(\xi) \equiv \bar{\phi}^j_{Y_1}(\xi)$ as defined in (43), and denote the true ChF of the SV process by $\hat{\phi}^j_1(\xi)$ as before. In first stage ($m = 1$) we have $\mathcal{E}^j_1(\xi) := \bar{\phi}^j_1(\xi) - \hat{\phi}^j_1(\xi)$ and

$$|\mathcal{E}^j_1(\xi)| \le |\hat{\phi}^j_1(\xi) - \phi^j_1(\xi)| + |\phi^j_1(\xi) - \bar{\phi}^j_1(\xi)| := |\hat{\mathcal{E}}^j_1(\xi)| + |\bar{\mathcal{E}}^j_1(\xi)|,$$

where $\phi^j_1(\xi)$ is the true ChF in the regime switching (RS) world (in the absence of projection error).[13] The first term, $\hat{\mathcal{E}}^j_1(\xi)$, is handled by Proposition 2. For the second term, with $\Phi_{j,k}(\xi) =$

[13] Note: $\hat{\mathcal{E}}^j_1(\xi)$ is error between regime switching (RS) approx. and true ChF, while $\bar{\mathcal{E}}^j_1(\xi)$ is error between true RS approx. and the RS approx. with PROJ error.

$P_{j,k}^{\Delta_t} \cdot \mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}^{m_0}))|G_{jk}]$, and $\bar{\Phi}_{j,k}(\xi)$ defined in (42),

$$\bar{\mathcal{E}}_1^j(\xi) = \sum_{k=1,..,m_0} \left(\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi)\right).$$

Denote by $p_{j,k}(x)$ the true conditional density, and $\bar{p}_{j,k}(x)$ the biorthogonal projection, where we factor out the probability $P_{j,k}^{\Delta_t}$ (whenever $P_{j,k}^{\Delta_t} > 0$). Then[14]

$$\begin{aligned}\frac{1}{P_{j,k}^{\Delta_t}}|\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi)| &\leq \int_{-\bar{\alpha}}^{\bar{\alpha}} |e^{\mathrm{i}\xi_n h(x)}(p_{j,k}(x) - \bar{p}_{j,k}(x))|dx + \tau_{j,k}(\bar{\alpha}) \\ &\leq \sqrt{2\bar{\alpha}}\, \|p_{j,k} - \bar{p}_{j,k}\|_2 + \tau_{j,k}(\bar{\alpha}) \leq C\sqrt{2\bar{\alpha}}\Delta^{p+1} + \tau(\bar{\alpha}),\end{aligned}$$

where $\tau_{j,k}(\bar{\alpha}) \leq \tau(\bar{\alpha}) := \max_{j,k} \tau_{j,k}(\bar{\alpha})$ is the density truncation error, which is controlled by choosing sufficiently large $\bar{\alpha}$. Here $\mathcal{O}(\Delta^{p+1})$ is the projection convergence rate (for the cubic basis $p = 3$), and C is a constant (see Kirkby (2015)). Hence

$$\max_{\xi_n} |\bar{\mathcal{E}}_1^j(\xi_n)| \leq \sum_{k=1,..,m_0} P_{j,k}^{\Delta_t} \left(C\sqrt{2\bar{\alpha}}\Delta^4 + \tau(\bar{\alpha})\right) = C_1\bar{\alpha}^{9/2}N^{-4} + \tau(\bar{\alpha}), \tag{46}$$

from our choice of $\Delta = 2\bar{\alpha}/(N-1)$. Thus, $|\mathcal{E}_1^j(\xi)| \leq A_1 m_0^{-d} + C_1\bar{\alpha}^{9/2}N^{-4} + \tau(\bar{\alpha})$ for some constant A_1. Now let $2 \leq m \leq M$, then

$$|\mathcal{E}_m^j(\xi)| \leq |\hat{\phi}_m^j(\xi) - \phi_m^j(\xi)| + |\phi_m^j(\xi) - \bar{\phi}_m^j(\xi)| := |\hat{\mathcal{E}}_m^j(\xi)| + |\bar{\mathcal{E}}_m^j(\xi)|.$$

The second term $\bar{\mathcal{E}}_m^j(\xi)$ now contains an additional error:

$$\bar{\mathcal{E}}_m^j(\xi) = \sum_{k=1,..,m_0} \left(\Phi_{j,k}(\xi)\phi_{m-1}^k(\xi) - \bar{\Phi}_{j,k}(\xi)\bar{\phi}_{m-1}^k(\xi)\right),$$

where we decompose $\bar{\mathcal{E}}_m^j(\xi) = \bar{\mathcal{E}}_{m,1}^j(\xi) + \bar{\mathcal{E}}_{m,2}^j(\xi)$ using

$$\begin{aligned}&\Phi_{j,k}(\xi)\phi_{m-1}^k(\xi) - \bar{\Phi}_{j,k}(\xi)\bar{\phi}_{m-1}^k(\xi) \\ &= \Phi_{j,k}(\xi)(\phi_{m-1}^k(\xi) - \bar{\phi}_{m-1}^k(\xi)) + \bar{\phi}_{m-1}^k(\xi)(\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi))\end{aligned}$$

For the first term, note that $|\Phi_{j,k}(\xi)| = P_{j,k}^{\Delta_t} \cdot |\mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}^{m_0}))|G_{jk}]| \leq P_{j,k}^{\Delta_t} \cdot \zeta_1(\xi)$, where $\zeta_1(\xi) = \max_{j,k} |\mathbb{E}[\exp(\mathrm{i}\xi h(X_{\Delta_t}^{m_0}))|G_{jk}]| \leq 1$, from which

$$\begin{aligned}|\bar{\mathcal{E}}_{m,1}^j(\xi)| &= \Big|\sum_{k=1,..,m_0} \Phi_{j,k}(\xi)(\phi_{m-1}^k(\xi) - \bar{\phi}_{m-1}^k(\xi))\Big| \\ &\leq \sum_{k=1,..,m_0} P_{j,k}^{\Delta_t}\zeta_1(\xi)|\phi_{m-1}^k(\xi) - \bar{\phi}_{m-1}^k(\xi)| \leq \zeta_1(\xi) \sum_{k=1,..,m_0} P_{j,k}^{\Delta_t}\bar{\mathcal{E}}_{m-1}^* = \zeta_1(\xi)\bar{\mathcal{E}}_{m-1}^*,\end{aligned}$$

where $\bar{\mathcal{E}}_m^* := \max_{1\leq j\leq m_0}\{\max_{\xi_n}\{|\bar{\mathcal{E}}_m^j(\xi_n)|\}\}$. Since from Remark 7 we may assume $|\bar{\phi}_m^k(\xi)| \leq 1$,

$$|\bar{\mathcal{E}}_{m,2}^j(\xi)| = \Big|\sum_{k=1,..,m_0} \bar{\phi}_{m-1}^k(\xi)(\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi))\Big| \leq \sum_{k=1,..,m_0} |\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi)| \leq \bar{\mathcal{E}}_1^*.$$

[14]Note that we are assuming the Gaussian quadrature error in the calculation of Ψ is dominated by the other sources (see Kirkby (2016) for a more detailed error analysis).

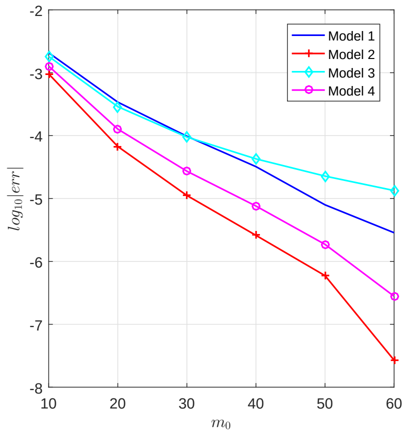

Figure 2: Convergence for swaps in Heston's model.

thus

$$\max_{1\leq j\leq m_0} |\bar{\mathcal{E}}^j_M(\xi_n)| \leq \bar{\mathcal{E}}^*_1 \cdot \sum_{m=0}^{M-1} \zeta_1(\xi_n)^m.$$

Also note that $\bar{\mathcal{E}}^j_M(\xi_1) = 0$ by design (i.e. the ChF at zero is one), and $\zeta_1(\xi_n) < 1$ for $n \geq 2$. Combining this with $\hat{\mathcal{E}}^j_M(\xi_n)$ from Proposition 2 (see (39)), the result follows. This completes the proof. $\square$

Remark 7 *For the second term, a weaker bound can also be found which justifies our assumption of boundedness on the term $\bar{\phi}^k_m(\xi)$. Note that*

$$\begin{aligned}|\bar{\phi}^k_{m-1}(\xi)(\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi))| &\leq \left(|\phi^k_{m-1}(\xi)| + |\bar{\phi}^k_{m-1}(\xi) - \phi^k_{m-1}(\xi)|\right)\left|\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi)\right| \\ &\leq \left(\zeta_2(\xi) + \bar{\mathcal{E}}^*_{m-1}\right)\left|\Phi_{j,k}(\xi) - \bar{\Phi}_{j,k}(\xi)\right|,\end{aligned}$$

where $\zeta_2(\xi) = \max_{1\leq m\leq M}\{\max_{1\leq k\leq m_0} |\phi^k_m(\xi)|\} \leq 1$. Hence $|\bar{\mathcal{E}}^j_{m,2}(\xi)| \leq (\zeta_2(\xi) + \bar{\mathcal{E}}^*_{m-1})\bar{\mathcal{E}}^*_1$. Combining the bound on $|\bar{\mathcal{E}}^j_{m,1}(\xi)|$ with this weaker bound on $|\bar{\mathcal{E}}^j_{m,2}(\xi)|$, we have $\bar{\mathcal{E}}^*_m \leq (\zeta_2(\xi) + \bar{\mathcal{E}}^*_{m-1})\bar{\mathcal{E}}^*_1 + \zeta_1(\xi)\bar{\mathcal{E}}^*_{m-1} \leq \zeta(\xi)\bar{\mathcal{E}}^*_1 + \bar{\mathcal{E}}^*_{m-1}(\zeta(\xi) + \bar{\mathcal{E}}^*_1)$, where $\zeta(\xi) = \max\{\zeta_1(\xi), \zeta_2(\xi)\} \leq 1$. Note that the error is damped by the term $\zeta(\xi) \to 0$ as $\xi \to \infty$. Setting $\zeta(\xi)$ to its max of one, we have the simplified bound $\bar{\mathcal{E}}^*_m \leq \bar{\mathcal{E}}^*_1 + \bar{\mathcal{E}}^*_{m-1}(1 + \bar{\mathcal{E}}^*_1)$. In particular,

$$\bar{\mathcal{E}}^*_M \leq \bar{\mathcal{E}}^*_1 \sum_{m=0}^{M-1} (1 + \bar{\mathcal{E}}^*_1)^m.$$

With $\bar{\alpha}$ chosen to control truncation error $\tau(\bar{\alpha}) << 1$, this term can be made arbitrarily small by in increasing N, where $\bar{\mathcal{E}}^*_1 \leq C_1\bar{\alpha}^{9/2}N^{-4} + \tau(\bar{\alpha})$ by (46). Hence we may assume a bound[15] *$|\bar{\phi}^k_m(\xi_n)| \leq 1$, where we note that $|\bar{\phi}^k_m(\xi_1)| = 1$ is enforced in the algorithm.*

5.1 Numerical convergence

As a numerical illustration of convergence, Figure 2 considers the CTMC error convergence for the variance swap values in Heston's model for which analytical prices are known via Bernard

[15] While we always have $|\phi^k_m(\xi)| \leq 1$, this ensures a bound on the approximated ChF.

Model	$\psi_L(\xi), \quad \omega = -\psi_L(-i)$	Fair Strike V_M
BSM	$\psi_L(\xi) = -\frac{\sigma^2}{2}\xi^2$ $\omega = -\frac{\sigma^2}{2}$	$\sigma^2 + \gamma^2\Delta_t$
MJD	$\psi_L(\xi) = -\frac{\sigma^2}{2}\xi^2 + \lambda\Big(\exp(i\xi\mu_J - \frac{\sigma_J^2}{2}\xi^2) - 1\Big)$ $\omega = -\frac{\sigma^2}{2} - \lambda\Big(\exp(\mu_J + \frac{\sigma_J^2}{2}) - 1\Big)$	$\sigma^2 + \lambda(\mu_J^2 + \sigma_J^2) + (\gamma + \lambda\mu_J)^2\Delta_t$
CGMY (KoBoL)	$\psi_L(\xi) = C\Gamma(-Y)\left((M - i\xi)^Y - M^Y + (G + i\xi)^Y - G^Y\right)$ $\omega = -C\Gamma(-Y)((M-1)^Y - M^Y + (G+1)^Y - G^Y)$	$C\Gamma(2-Y)(M^{Y-2} + G^{Y-2})$ $+ \left(\gamma + C\Gamma(1-Y)(M^{Y-1} - G^{Y-1})\right)^2\Delta_t$
NIG	$\psi_L(\xi) = -\delta\left(\sqrt{\alpha^2 - (\beta + i\xi)^2} - \sqrt{\alpha^2 - \beta^2}\right)$ $\omega = \delta\left(\sqrt{\alpha^2 - (\beta+1)^2} - \sqrt{\alpha^2 - \beta^2}\right)$	$\delta\alpha^2(\alpha^2 - \beta^2)^{-3/2} + \left(\gamma + \delta\beta/\sqrt{\alpha^2 - \beta^2}\right)^2\Delta_t$
Kou (DE)	$\psi_L(\xi) = -\frac{\sigma^2}{2}\xi^2 + \lambda\Big(\frac{p\eta_1}{\eta_1 - i\xi} + \frac{(1-p)\eta_2}{\eta_2 + i\xi} - 1\Big)$ $\omega = -\frac{\sigma^2}{2} - \lambda\Big(\frac{p\eta_1}{\eta_1 - 1} + \frac{(1-p)\eta_2}{\eta_2 + 1} - 1\Big)$	$\sigma^2 + 2\frac{\lambda p}{\eta_1^2} + 2\frac{\lambda(1-p)}{\eta_2^2} + \left(\gamma + \frac{\lambda p}{\eta_1} - \frac{\lambda(1-p)}{\eta_2}\right)^2\Delta_t$
VG	$\psi_L(\xi) = -\frac{\sigma^2}{2}\xi^2 - \frac{1}{\nu}\log\left(1 - i\nu\theta\xi + \nu\frac{\sigma_V^2}{2}\xi^2\right)$ $\omega = -\frac{\sigma^2}{2} + \frac{1}{\nu}\log\left(1 - \nu\theta - \nu\frac{\sigma_V^2}{2}\right)$	$\sigma^2 + \sigma_V^2 + \nu\theta^2 + (\gamma + \theta)^2\Delta_t$

Figure 3: Levy symbols $\psi_L(\xi)$, and analytic variance swap fair strike, where $\gamma := r - q - \psi_L(-i) = r - q + \omega$.

and Cui (2014). Four models are considered with parameters $r = 0.05, T = 1, M = 52$ and model parameters ordered respectively as $\kappa \in \{6.5, 2.5, 1.5, 3\}$, $v_0 \in \{0.03, 0.04, 0.045, 0.03\}$, $\theta \in \{0.035, 0.03, 0.035, 0.045\}$, $\sigma_v \in \{0.45, 0.20, 0.25, 0.30\}$, and $\rho \in \{-0.7, -0.7, -0.8, -0.6\}$. In each case, we fix $N = 2^{10}$ to isolate the convergence in m_0. In practice, we find that a value of $m_0 = 40$ tends to be sufficient for basis point accuracy, but given a large fixed computational budget, increased accuracy is achievable. The algebraic convergence in m_0 from Proposition 2 and Proposition 3 is apparent in Figure 2.

6 Numerical examples

6.1 An Experiment with Lévy Models

Before investigating the performance of the proposed methodology for SV models, we first consider the case of an exponential Lévy process for which the price of a discrete variance swap is available in closed-form. In particular, suppose that $X_m = \ln(S_m/S_{m-1})$ is the discrete return of a risk-neutral Lévy model. The fair strike of a discrete variance swap satisfies

$$V_M = \frac{1}{T}\mathbb{E}_0\Big[\sum_{m=1}^{M} X_m^2\Big] = \frac{M}{T}\mathbb{E}[X_1^2] = c_2 + c_1^2\Delta_t,$$

where c_i denotes the ith cumulant of $\ln(S_{t+1}/S_t)$. Table 3 provides the fair price[16] for several common Lévy models, as well as the characteristic exponent $\psi_L(\xi)$ used for pricing. In particular, the risk-neutral characteristic function of $X_{\Delta_t} = \ln(S_{t+\Delta_t}/S_t)$ is given by $\exp((r - q - \psi_L(-i))\Delta_t + \psi_L(\xi)\Delta_t)$.

[16]For the Mixed normal jump diffusion (MNJD), the characteristic exponent is given as a weighted combination of $\psi_L(\xi)$ for two MJD models, with weights p and $1-p$. See for example Bhat and Kumar (2012), who use this model to approximate a Markov tree model.

By independence and stationarity of Lévy increments, the ChF of $Y_M := T \cdot V_M$ is given by $\phi_{Y_M}(\xi) = (\mathbb{E}[\exp(i\xi X_1^2)])^M$, and the same methodology can be applied as before, in the special case of a single state.[17]

M	BSM	MJD	Kou(DE)	NIG	CGMY	MNJD
12	1.29e-10	2.85e-08	1.97e-05	1.27e-08	9.36e-07	8.15e-08
24	7.94e-12	3.93e-08	9.64e-06	1.95e-08	8.12e-07	4.66e-09
52	4.88e-13	8.89e-08	2.17e-06	5.44e-09	7.57e-07	8.45e-07
100	2.64e-13	3.76e-07	1.27e-06	1.64e-08	7.33e-07	2.01e-07
252	5.78e-11	1.45e-06	3.59e-06	2.22e-08	7.18e-07	2.10e-06
500	3.47e-10	2.10e-06	4.35e-06	2.37e-08	7.13e-07	3.59e-06
10000	1.62e-08	2.97e-06	3.71e-06	2.47e-08	7.07e-07	5.41e-06

Table 4: Variance swap experiment with Lévy processes, $T = 1$, $r = 0.05$, $q = 0$. $N = 2^{10}$. α chosen by cumulant rule with $L_1 = 14$. BSM $\sigma = 0.18$, MJD $(\sigma, \lambda, \mu_J, \sigma_J) = (0.12, 1, -0.12, 0.08)$, Kou $(\sigma, \lambda, p, \eta_1, \eta_2) =$ (0.08, 1, 0.4, 10, 5), NIG $(\alpha, \beta, \delta) =$ (15, -5, 0.5), CGMY $= (1, 5, 5, 0.5)$, MNJD $(\sigma, \lambda, p, a_1, b_1, a_2, b_2) =$(0.18, 1, 0.6, -0.12, 0.12, 0.15, 0.08).

Table 4 compares the errors for several Lévy models: Black-Scholes-Merton (BSM), Merton's jump diffusion (MJD), Kou's Double Exponential jump diffusion (Kou(DE)), Normal Inverse Gaussian (NIG), KoBoL (CGMY), Mixed Normal Jump Diffusion (MNJD). For these models, we use the same PROJ parameters, $L_1 = 14$ and $N = 2^{10}$, as for all experiments with stochastic volatility. In particular, note that the error is very stable with respect to M, ranging from $M = 12$ to $M = 10000$. While greater accuracy is obtainable by increasing the number of basis elements N, we find that $N = 2^{10}$ is sufficient for SV models, in the sense that the overall error in that case will be dominated by the error of the Markov chain approximation to the variance process. The cpu cost for variance swaps and options is 0.21 seconds with $N = 2^{10}$, and 0.06 seconds with $N = 2^9$, with similar accuracy (note that the same algorithm is used to price swaps and options).[18]

[17]Note that this gives a method for pricing options on realized variance for Lévy models, which do not have a simple closed-form expression.

[18]In most cases, a value of $N = 2^9$ is more than sufficient for practical accuracy (e-06 or better), while $N = 2^{10}$ is conservative. Similarly, a value of $L_1 = 10$ is generally sufficient, but for CGMY, $L_1 = 14$ is preferred. Rather than make L_1 model dependent, we have chosen the more conservative value in the experiments.

K	BSM	MJD	Kou(DE)	NIG	CGMY	MNJD
(Swap)	0.032405	0.035204	0.062406	0.039778	0.158536	0.061242
0.01	0.021312	0.023975	0.051346	0.028335	0.141292	0.048742
0.02	0.011800	0.016777	0.047335	0.019626	0.131822	0.039230
0.03	0.002579	0.012444	0.044056	0.013340	0.122545	0.029838
0.04	0.000007	0.009132	0.041234	0.009277	0.113640	0.023479
0.05	0.000000	0.006640	0.038756	0.006660	0.105234	0.019073

Table 5: Variance call options with Lévy processes, $T = 1$, $r = 0.05$, $q = 0$, $M = 252$. PROJ: $N = 2^{10}$, $\bar{\alpha}$ chosen by cumulant rule with $L_1 = 14$. Same model parameters as in Table 4.

While analytical formulas are available for variance swaps under Lévy models, values for call options on realized variance are still unknown. Hence, we provide a set of reference prices in Table 5 obtained by PROJ. Here we clearly see the effect of tail heaviness on increasing strike. For the BSM model, fast decay of the log return tails leads to rapidly decaying prices OTM. However, prices remain relatively high at each strike for the models with jumps, especially for the CGMY model. A similar observation holds for MJD, for which the initial swap prices are similar to BSM but decay slowly OTM.

6.2 Heston's model

In this section, we consider the problem of pricing discrete variance swaps in Heston's model using the proposed method. In subsequent sections, more general stochastic volatility models are considered. We find that the following parameter settings produce good results, obtaining at least a practical level of $1e$-04 accuracy or better: $N = 2^{10}, L_1 = 14$, $m_0 = 40$. The parameter γ is listed in Table 11 for each model class, though in general a value of $\gamma = 4.5 \sim 6.5$ works well (for $\alpha-$H and Hull-White, a value of 8.5 produces modestly better results). For the PROJ method, CPU times in seconds for $M = \{5, 12, 54, 180, 360\}$ are respectively $\{2.43, 2.50, 2.70, 3.43, 4.59\}$.

	$\rho = -0.7$						$\rho = -0.1$					
M	PROJ	BC	MC	Std.Err	E_{PROJ}	E_{MC}	PROJ	BC	MC	Std.Err	E_{PROJ}	E_{MC}
5	0.017978	0.017960	0.017885	1.40e-05	1.85e-05	7.43e-05	0.017738	0.017737	0.017711	1.38e-05	2.40e-07	2.68e-05
12	0.017788	0.017770	0.017728	9.65e-06	1.83e-05	4.22e-05	0.017655	0.017655	0.017618	1.01e-05	4.17e-07	3.67e-05
54	0.017652	0.017634	0.017595	5.89e-06	1.81e-05	3.87e-05	0.017605	0.017604	0.017566	6.96e-06	4.01e-07	3.84e-05
180	0.017621	0.017603	0.017569	4.80e-06	1.81e-05	3.40e-05	0.017594	0.017594	0.017563	6.10e-06	3.91e-07	3.10e-05
360	0.017614	0.017596	0.017563	4.53e-06	1.81e-05	3.35e-05	0.017592	0.017592	0.017562	5.90e-06	3.82e-07	2.93e-05

Table 6: Discrete Variance Swap under Heston's model (without jumps). Parameters: $r = 0.0319, V_0 = (10.11\%)^2, \kappa = 6.21, \theta = 0.019, \sigma_v = 0.31, T = 1$. PROJ values obtained using $N = 2^{10}, L_1 = 14, m_0 = 40$, $\gamma = 5.5$. $E_{PROJ} := |PROJ - BC|, E_{MC} := |MC - BC|$.

	$\rho = -0.7$						$\rho = -0.1$					
M	PROJ	BC	MC	Std.Err	E_{PROJ}	E_{MC}	PROJ	BC	MC	Std.Err	E_{PROJ}	E_{MC}
5	0.037574	0.037574	0.037434	2.79e-05	1.65e-07	1.40e-04	0.037120	0.037121	0.037045	2.78e-05	6.95e-08	7.57e-05
12	0.037169	0.037169	0.037098	1.92e-05	1.97e-07	7.01e-05	0.036957	0.036957	0.036912	2.01e-05	8.00e-09	4.52e-05
54	0.036911	0.036911	0.036860	1.19e-05	1.88e-07	5.09e-05	0.036861	0.036861	0.036802	1.39e-05	5.95e-09	5.87e-05
180	0.036857	0.036856	0.036777	9.78e-06	1.86e-07	7.99e-05	0.036841	0.036841	0.036778	1.22e-05	2.56e-09	6.33e-05
360	0.036845	0.036845	0.036785	9.27e-06	1.86e-07	5.95e-05	0.036837	0.036837	0.036772	1.18e-05	2.91e-09	6.47e-05

Table 7: Discrete Variance Swap under Heston's model (without jumps). Parameters: $r = 0.05, V_0 = 0.03, \kappa = 3, \theta = 0.04, \sigma_v = 0.25, T = 1$. PROJ values obtained using $N = 2^{10}, L_1 = 14, m_0 = 40$, $\gamma = 5.5$. $E_{PROJ} := |PROJ - BC|, E_{MC} := |MC - BC|$.

For the first set of stochastic volatility experiments, we obtain the price of discrete variance swaps for which analytical reference prices are available using the closed-form formulae in Bernard and Cui (2014), reported in columns "BC" in Tables 6 and 7, while ours are reported in "PROJ" columns. Prices and errors are reported as a function of M, the discrete monitoring frequency, and for low and high negative correlation: $\rho = -0.1$ and $\rho = -0.7$. While analytical results are available for Heston's model, we present Monte Carlo (MC) method as well to establish a baseline for cases without reference prices (see Section 6.2.1 for details). As with the Lévy experiments in Section 6.1, errors are fairly invariant with respect to M. Also note that in the first model, errors for MC range from 2.68e-05 $\sim$ 7.43e-05, while errors for PROJ range from 2.40e-07 $\sim$ 1.85e-05. In the second model, errors for MC range from 4.52e-05 $\sim$ 1.40e-04, while for PROJ they range from 2.91e-09 $\sim$ 1.97e-07.

	$\rho = -0.7$				$\rho = -0.1$			
K	PROJ	MC	Std.Err.	Diff	PROJ	MC	Std.Err.	Diff
0.01	0.025602	0.025605	6.94e-07	3.14e-06	0.025552	0.025555	6.83e-07	2.62e-06
0.02	0.016278	0.016274	1.06e-06	4.21e-06	0.016247	0.016252	1.12e-06	5.57e-06
0.03	0.008700	0.008682	2.91e-06	1.82e-05	0.008650	0.008661	2.95e-06	1.07e-05
0.04	0.004100	0.004085	4.09e-06	1.50e-05	0.003993	0.003997	4.02e-06	3.69e-06
0.05	0.001803	0.001793	3.95e-06	1.03e-05	0.001683	0.001692	3.78e-06	8.69e-06

Table 8: Variance call option under Heston's model (without jumps). Parameters: $M = 52, r = 0.05, V_0 = 0.03, \kappa = 3, \theta = 0.04, \sigma_v = 0.25, T = 1$. PROJ values obtained using $N = 2^{10}$, $L_1 = 14, m_0 = 40$, $\gamma = 5.5$. Monte Carlo with analytical swap price as control variate.

In Table 8, we consider variance call options under Heston's model without jumps. To validate our method, we also conduct Monte Carlo simulations using the Euler scheme of 6.2.1, but with the addition of analytical swap prices as a control variate to reduce variance. The simulation is conducted using 10^6 sample paths. We report the prices for the projection method and Monte Carlo as well as the absolute difference (Diff) between the two methods. While the difference is very small (2.62e-06 $\sim$ 1.82e-05), the comparison with analytical swap values for this model in Table 7 suggests that the error for PROJ is smaller than estimated.

	$\rho = -0.7$				$\rho = -0.1$			
K	PROJ	MC	Std.Err.	Diff	PROJ	MC	Std.Err.	Diff
0.01	0.056504	0.056322	4.57e-05	1.82e-04	0.056454	0.056307	4.88e-05	1.47e-04
0.02	0.047057	0.046937	4.57e-05	1.20e-04	0.047027	0.046885	4.88e-05	1.42e-04
0.03	0.038418	0.038402	4.55e-05	1.59e-05	0.038419	0.038307	4.81e-05	1.12e-04
0.04	0.031475	0.031418	4.46e-05	5.68e-05	0.031458	0.031362	4.68e-05	9.52e-05
0.05	0.026155	0.026088	4.31e-05	6.68e-05	0.026111	0.026020	4.46e-05	9.09e-05

Table 9: Variance call option under Heston's model with normal jumps. Parameters: $M = 52, r = 0.05, V_0 = 0.03, \kappa = 3, \theta = 0.04, \sigma_v = 0.25, T = 1$. Jump pars: $a_1 = -0.10, b_1 = 0.15, \lambda = 1$. PROJ values obtained using $N = 2^{10}$, $L_1 = 14, m_0 = 40, \gamma = 5.5$.

Table 9 introduces normal jumps in the log-return process (Bate's model), with the same stochastic volatility parameters as in the previous example. While the difference with MC is now larger, the discrepancy is likely attributable to the increased variance arising from jumps in the simulation, with standard errors increasing more than ten fold by incorporating jumps. This highlights one of the benefits of our algorithm over MC, which requires increased computational effort to handle jumps. It is evident from Tables 8 and 9 that jumps can have a significant impact on the value of variance call options, which confirms the findings of Broadie and Jain (2008), among others. As we observed for Lévy models, the price decay OTM is much slower in the presence of jumps, reflecting the fatter tail of realized variance. Since returns are squared, the impact of jumps is amplified.

6.2.1 Monte Carlo

For each Monte Carlo simulation in this work, we apply an Euler scheme with time interval partitioning to reduce discretization bias. In particular, for a fixed interval $[0, T]$, let $\{t_m\}_{m=0}^{M}$ denote the actual discrete monitoring points, and let $\{t_j\}_{j=0}^{\bar{M}}$ denote a refined partition, where $t_j = j\delta_t$, $\delta_t = T/\bar{M}$, and $\bar{M} := b \cdot M$ for some $b \in \mathbb{N}_+$. For each of $\bar{N}$ sample paths, we simulate the level of variance $\{v_j^{(i)}\}_{j=0}^{\bar{M}}$ by[19]

$$v_{j+1}^{(i)} = v_j^{(i)} + \hat{\mu}(v_j^{(i)})\delta_t + \hat{\sigma}(v_j^{(i)})\sqrt{\delta_t} \cdot W_{2,j}^{(i)},$$

as well as the log asset process $Z_t = \log(S_t)$:

$$Z_{j+1}^{(i)} = Z_j^{(i)} + \left(\gamma - \frac{1}{2}m^2(v_j^{(i)})\right)\delta_t + m(v_j^{(i)})\sqrt{\delta_t}\left(\rho W_{2,j}^{(i)} + \sqrt{1-\rho^2}W_{*,j}^{(i)}\right) + \sum_{k=1}^{N_{\delta_t}} Y_k$$

[19]For variance processes that require positivity, we take $v_{j+1}^{(i)} = \max\{0, v_{j+1}^{(i)}\}$. This is known as the *absorption scheme* (c.f. Lord et al. (2010)). Alternative methods such as reflection were also tested, with similar results, although we find as in Lord et al. (2010) that absorption leads to less bias than reflection.

where $W_{2,j}^{(i)}$ and $W_{*,j}^{(i)}$ are independent standard normal random variables (see Broadie and Jain (2008) for an Euler Scheme on the level of S_t). The realized variance is then defined by (1), by summing over returns defined by the actual monitoring points $\{t_m\}_{m=0}^{M}$. For numerical experiments we set $\bar{N} = 10^6$, which we find to be sufficient to achieve a standard error on the order $e-05$ or better for the examples considered, with a goal of basis point accuracy.[20]

The use of the Euler scheme is for its simplicity and applicability to the models we can consider with the proposed methodology. For *specific* models, such as Heston, exact simulation schemes exist to eliminate the time integration bias (Broadie and Kaya (2006)). The burden of employing an exact scheme for each of the seven methods, if this were even possible, eliminates the main advantage of Monte Carlo over the proposed methodology, which is its simplicity. From an implementation perspective, the addition of a new model within our framework requires just several lines of code (less even than a new Euler scheme), while implementing a sophisticated and model-tailored Monte Carlo scheme becomes much more complex.

MC: $(\bar{N}, b)$	$(10^4, 3)$	$(10^5, 3)$	$(10^5, 5)$	$(10^6, 5)$	$(10^6, 10)$
Err	1.64e-03	1.16e-03	8.91e-04	9.41e-04	7.41e-04
Std. Err	3.37e-04	1.06e-04	1.06e-04	3.35e-05	3.36e-05
CPU	0.16	1.28	1.75	21.78	37.32
Jump-CPU	0.39	3.52	4.63	52.19	89.40
PROJ: (m_0, N)	$(10, 2^9)$	$(20, 2^9)$	$(30, 2^9)$	$(40, 2^9)$	$(40, 2^{10})$
Err	9.24e-04	2.81e-04	1.27e-04	5.70e-05	5.71e-05
CPU	0.31	0.52	0.80	1.08	2.98
Jump-CPU	0.31	0.52	0.80	1.08	2.98

Table 10: Monte Carlo vs PROJ. Variance swap in Heston's model with $M = 100, T = 2, r = 0.05, \eta = 0.5, \theta = 0.03, \rho = -0.5, \sigma_v = 0.35, v_0 = 0.025$. Jump-CPU is the cost of algorithm upon adding a double exponential jump component.

A comparison between PROJ and the simple Euler MC scheme is provided in Table 10 for a variance swap in Heston's model with analytic price of 0.02692212, found using the method of Bernard and Cui (2014). The two parameters for MC are the number of paths $\bar{N}$, and the refinement parameter b. While analytic prices for the jump case are unknown, we also illustrate the major CPU cost increase for MC when a double exponential jump component is added (labeled "Jump-CPU"), compared with PROJ, which is unaffected by an arbitrary jump component.[21] Moreover, the PROJ method easily handles infinite activity jump processes,

[20]The value of b is also chosen with this goal in mind, in order to reduce bias. The appropriate level of b can vary by model, and we use trial and error to determine an appropriate value for the experiments.

[21]Note that the final PROJ column with $(m_0, N) = (40, 2^{10})$ is the parameter setting for all other experiments. The choice of $N = 2^{10}$ is conservative for basis point accuracy. If speed is the primary concern, a value of $N = 2^9$ is generally sufficient.

which present a challenge for MC, in terms of cost, accuracy, and implementation. Note that while various modifications may be added to reduce the variance of the simple MC scheme, eliminating the bias in a generic way is challenging. We find that for certain parameter settings (especially for large variance levels), the MC bias may not be eliminated, resulting in large errors.

6.3 Additional Stochastic Volatility Models

In this section, we provide numerical results for several additional SV models including Stein-Stein, Hull-White, 3/2, 4/2, Scott, and α-Hypergeometric models. Two test sets are provided in Table 11 for the stochastic volatility parameters. All numerical results for these models are new to the literature. For each model, results are reported without jumps, as well as for three jump models: double exponential, normal, and mixed normal, with common jump parameters listed in Table 12.

	Test Set 1						Test Set 2						
Model	σ_v	η	ρ	θ	v_0	a_v	σ_v	η	ρ	θ	v_0	a_v	γ
SS	0.20	2	-0.5	0.18	0.19	-	0.15	5	-0.7	0.18	0.15	-	5.5
HW	0.30	-	-0.7	-	0.035	0.01	0.25	-	-0.6	-	0.03	0.01	8.5
3/2	0.15	5	-0.7	0.03	0.03	-	0.12	2	-0.3	0.04	0.035	-	5.5
4/2	0.20	4	-0.7	0.035	0.04	-	0.15	3	-0.5	0.04	0.045	-	5.5
Hes	0.35	6.5	-0.75	0.03	0.03	-	0.15	4	-0.7	0.035	0.03	-	5.5
Scott	0.15	3	-0.6	log(0.15)	log(0.15)	-	0.20	2	-0.9	log(0.16)	log(0.18)	-	5.5
$\alpha-$H	0.15	0.01	-0.6	0.04	log(0.19)	0.01	0.20	0.05	-0.9	0.2	log(0.17)	0.03	8.5

Table 11: Parameters for SV tests. In all cases, $r = 0.05$, $S_0 = 1$. For the 4/2 model: $a_S = 0.5, b_S = 0.5 \cdot v_0$.

Jump Model	Parameters
Double Exponential	$\eta_1 = 15, \eta_2 = 10, p = 0.40, \lambda = 1$
Normal	$a_1 = -0.10, b_1 = 0.15, \lambda = 1$
Mixed Normals	$a_1 = -0.12, b_1 = 0.12, a_2 = 0.15, b_2 = 0.08, p = 0.6, \lambda = 1$

Table 12: Jump distribution parameters for numerical experiments

The first set of experiments is presented in Table 13 for each model in the absence of jumps, with $T = 0.5$ and $M = 52$. Estimated swap prices are provided for PROJ and Monte Carlo, along with the absolute difference (Diff) between the two estimates. As before, we partition the time space for MC to reduce bias. Normal jumps are introduced in Table 14, and again we notice the increased standard errors from MC. Finally, Table 15 presents results for the model with double exponential jumps.

	Test Set 1				Test Set 2			
Model	PROJ	MC	Std.Err.	Diff	PROJ	MC	Std.Err.	Diff
SS	0.040385	0.040431	2.27e-05	4.60e-05	0.030473	0.030402	1.07e-05	7.07e-05
HW	0.035117	0.035038	7.89e-06	7.93e-05	0.030097	0.030046	6.56e-06	5.07e-05
3/2	0.030012	0.029975	5.87e-06	3.73e-05	0.035110	0.035047	6.86e-06	6.26e-05
4/2	0.041447	0.041349	8.36e-06	9.80e-05	0.045875	0.045795	9.07e-06	7.94e-05
Hes	0.030061	0.029995	1.04e-05	6.62e-05	0.032866	0.032799	8.26e-06	6.68e-05
Scott	0.022633	0.022599	4.71e-06	3.37e-05	0.030053	0.030054	6.50e-06	5.72e-07
$\alpha-$H	0.035995	0.035933	8.17e-06	6.23e-05	0.027438	0.027481	6.47e-06	4.23e-05

Table 13: PROJ vs MC for variance swaps with $T = 0.5$, $M = 52$ and no jumps. MC paths $= 10^6$, and partitioning time steps in path generation to reduce bias. For Hull-White and Heston, analytical values used for CV. PROJ $N = 2^{10}$, $m_0 = 40$, $L_1 = 14$, nonuniform grid multiplier $= 4/5$.

	Test Set 1				Test Set 2			
Model	PROJ	MC	Std. Err.	Diff	PROJ	MC	Std. Err.	Diff
SS	0.072877	0.072866	7.26e-05	1.13e-05	0.062970	0.062925	6.95e-05	4.44e-05
HW	0.067600	0.067478	7.09e-05	1.22e-04	0.062575	0.062472	7.02e-05	1.02e-04
3/2	0.062510	0.062384	7.05e-05	1.27e-04	0.067592	0.067550	7.16e-05	4.13e-05
4/2	0.073941	0.073930	7.28e-05	1.16e-05	0.078373	0.078367	7.33e-05	6.10e-06
Hes	0.062559	0.062523	6.92e-05	3.60e-05	0.065347	0.065302	7.04e-05	4.52e-05
Scott	0.055138	0.055096	6.92e-05	4.19e-05	0.062532	0.062525	6.99e-05	6.99e-06
$\alpha-$H	0.068478	0.068262	7.09e-05	2.16e-04	0.059934	0.059755	6.88e-05	1.79e-04

Table 14: PROJ vs MC for variance swaps with $T = 0.5$, $M = 52$ and normal jumps: $a = -0.10, b = 0.15, \lambda = 1$. MC paths $= 10^6$, and partitioning time steps in path generation to reduce bias. PROJ $N = 2^{10}$, $m_0 = 40$, $L_1 = 14$, nonuniform grid multiplier $= 4/5$.

Additional variance swap reference prices obtained by the PROJ method are provided in Table 16 for the full set of models and each jump distribution, with monitoring frequencies $M \in \{12, 24, 52, 100, 252\}$. For each M, the CPU time in seconds for PROJ is $\{2.45, 2.52, 2.68, 2.96, 3.91\}$, for which the main cost is the recursive algorithm for $\phi_{Y_M}(\xi)$. However, once $\phi_{Y_M}(\xi)$ is obtained, we can price swaps as well as an entire spectrum of call options on variance at a negligible additional cost (equivalent to the cost of pricing a set of vanilla European options). Moreover, the same $\phi_{Y_M}(\xi)$ can be used to price a full spectrum of volatility swaps and options. For the

	Test Set 1				Test Set 2			
Model	PROJ	MC	Std. Err.	Diff	PROJ	MC	Std. Err.	Diff
SS	0.055936	0.055941	5.89e-05	5.31e-06	0.046025	0.045866	5.59e-05	1.59e-04
HW	0.050666	0.050580	5.64e-05	8.53e-05	0.045638	0.045575	5.59e-05	6.33e-05
3/2	0.045565	0.045511	5.64e-05	5.41e-05	0.050661	0.050626	5.69e-05	3.56e-05
4/2	0.057002	0.056874	5.65e-05	1.28e-04	0.061429	0.061400	5.74e-05	2.86e-05
Hes	0.045612	0.045489	5.56e-05	1.24e-04	0.048411	0.048304	5.61e-05	1.07e-04
Scott	0.038171	0.038130	5.57e-05	4.10e-05	0.045595	0.045593	5.62e-05	2.43e-06
$\alpha-$H	0.051544	0.051480	5.63e-05	6.40e-05	0.042986	0.042971	5.52e-05	1.44e-05

Table 15: PROJ vs MC for variance swaps with $T = 0.5$, $M = 52$ and DE jumps: $\eta_1 = 15, \eta_2 = 10, p = 0.40, \lambda = 1$. MC paths $= 10^6$, and partitioning time steps in path generation to reduce bias. PROJ $N = 2^{10}$, $m_0 = 40$, $L_1 = 14$, nonuniform grid multiplier $= 4/5$.

Monte Carlo method, which is the only method with comparable generality, the computational cost is measured in minutes, rather than a few seconds. We also provide reference prices for variance call options in Table 17 for the full set of models and each jump distribution.

7 Conclusion

By blending a continuous-time Markov chain approximation for the variance process, a regime-switching approximation for the underlying, and the Fourier method of frame projection (PROJ), we develop a novel and highly efficient and accurate transform-based method to price swaps and options related to discretely-sampled realized variance/volatility under a general class of stochastic volatility models with jumps. Accurate prices are obtained in just a few seconds, compared with minutes for Monte Carlo, which is the only existing method with comparable generality. Despite the use of Fourier techniques, our approach does not require the characteristic function of transition densities or integrated variance, which are not available for many models in practice. The method applies to discrete variance and volatility swaps and options with log or relative returns. Models considered include several popular stochastic volatility models augmented by jumps: Heston, Scott, Hull-White, Stein-Stein, α-Hypergeometric, 3/2 and 4/2 models. Numerical experiments confirm the accuracy of our method. Interesting directions for future work include extensions to stochastic volatility models with jumps in both the asset and variance as in Zheng and Kwok (2014), the three factor model of Pun et al. (2015) and the mean reverting stochastic volatility model of Wong and Lo (2009), or the class of time-changed diffusion models as in Li et al. (2016). Given the recursive structure of the algorithm, extensions to geometric Asian options, cliquets and equity indexed annuities are also promising.

	Test Set 1					Test Set 2				
M	12	24	52	100	252	12	24	52	100	252
	No Jumps									
SS	0.040551	0.040444	0.040385	0.040361	0.040345	0.030607	0.030521	0.030473	0.030453	0.030440
HW	0.035174	0.035137	0.035117	0.035109	0.035104	0.030149	0.030115	0.030097	0.030089	0.030084
3/2	0.030053	0.030026	0.030012	0.030006	0.030003	0.035144	0.035122	0.035110	0.035105	0.035101
4/2	0.041466	0.041454	0.041447	0.041444	0.041442	0.045895	0.045882	0.045875	0.045872	0.045870
Hes	0.030215	0.030117	0.030061	0.030038	0.030022	0.032954	0.032897	0.032866	0.032853	0.032844
Scott	0.022690	0.022653	0.022633	0.022624	0.022620	0.030122	0.030077	0.030053	0.030043	0.030037
α-H	0.036048	0.036013	0.035995	0.035987	0.035982	0.027507	0.027462	0.027438	0.027428	0.027422
	Normal Jumps									
SS	0.073020	0.072928	0.072877	0.072856	0.072842	0.063080	0.063012	0.062970	0.062951	0.062939
HW	0.067645	0.067618	0.067600	0.067594	0.067590	0.062619	0.062596	0.062575	0.062563	0.062553
3/2	0.062522	0.062516	0.062510	0.062508	0.062505	0.067614	0.067601	0.067592	0.067589	0.067590
4/2	0.073939	0.073937	0.073941	0.073949	0.073959	0.078370	0.078368	0.078373	0.078379	0.078385
Hes	0.062690	0.062609	0.062559	0.062537	0.062523	0.065424	0.065378	0.065347	0.065334	0.065326
Scott	0.055162	0.055151	0.055138	0.055129	0.055121	0.062592	0.062558	0.062532	0.062519	0.062509
α-H	0.068519	0.068494	0.068478	0.068472	0.068470	0.059978	0.059952	0.059934	0.059926	0.059919
	Double Exponential Jumps									
SS	0.056088	0.055990	0.055936	0.055913	0.055899	0.046148	0.046070	0.046025	0.046005	0.045993
HW	0.050712	0.050681	0.050666	0.050662	0.050660	0.045687	0.045657	0.045638	0.045632	0.045629
3/2	0.045592	0.045576	0.045565	0.045561	0.045557	0.050682	0.050666	0.050661	0.050665	0.050671
4/2	0.057006	0.057000	0.057002	0.057004	0.057004	0.061436	0.061430	0.061429	0.061426	0.061420
Hes	0.045757	0.045666	0.045612	0.045589	0.045573	0.048492	0.048440	0.048411	0.048401	0.048395
Scott	0.038231	0.038200	0.038171	0.038153	0.038138	0.045660	0.045619	0.045595	0.045586	0.045581
α-H	0.051586	0.051557	0.051544	0.051542	0.051542	0.043047	0.043010	0.042986	0.042975	0.042968
	Mixture of Normal Jumps									
SS	0.069360	0.069268	0.069217	0.069196	0.069182	0.059419	0.059349	0.059307	0.059289	0.059278
HW	0.063985	0.063959	0.063945	0.063941	0.063940	0.058958	0.058935	0.058918	0.058910	0.058905
3/2	0.058863	0.058854	0.058848	0.058845	0.058843	0.063955	0.063943	0.063939	0.063942	0.063948
4/2	0.070280	0.070279	0.070284	0.070289	0.070295	0.074711	0.074710	0.074712	0.074713	0.074711
Hes	0.059028	0.058946	0.058896	0.058874	0.058859	0.061765	0.061719	0.061691	0.061680	0.061674
Scott	0.051501	0.051482	0.051461	0.051447	0.051435	0.058932	0.058898	0.058875	0.058865	0.058858
α-H	0.064859	0.064836	0.064823	0.064821	0.064821	0.056318	0.056289	0.056270	0.056261	0.056254

Table 16: Swaps with SV parameters from Table 11 and jump distribution parameters from Table 12. Other parameters: $T = 0.5$, $r = 0.05$. Prices found with $N = 2^{10}$, $m_0 = 40$, and $L_1 = 14$. CPU times in sec for $M = \{12, 24, 52, 100, 252\}$ are respectively $\{2.45, 2.52, 2.68, 2.96, 3.91\}$

	Test Set 1					Test Set 2				
K	0.01	0.02	0.03	0.04	0.05	0.01	0.02	0.03	0.04	0.05
	No Jumps: $T = 1, M = 100$									
SS	0.029964	0.021570	0.014808	0.009786	0.006278	0.021326	0.012083	0.005045	0.001583	0.000405
HW	0.023982	0.014488	0.006124	0.001711	0.000366	0.019191	0.009730	0.002567	0.000350	0.000034
3/2	0.019036	0.009760	0.003151	0.000641	0.000095	0.023966	0.014454	0.005322	0.000601	0.000019
4/2	0.030203	0.020690	0.011212	0.003298	0.000379	0.034373	0.024861	0.015353	0.006378	0.001187
Hes	0.019077	0.009879	0.003595	0.001032	0.000261	0.022640	0.013143	0.004839	0.000980	0.000130
Scott	0.012041	0.002954	0.000062	0.000000	0.000000	0.017883	0.008439	0.001765	0.000159	0.000009
α-H	0.024618	0.015115	0.006578	0.001973	0.000494	0.015248	0.006549	0.001994	0.000559	0.000163
	Normal Jumps: $T = 0.5, M = 50$									
SS	0.061403	0.052383	0.044626	0.038272	0.033181	0.051661	0.042364	0.035018	0.030025	0.026530
HW	0.056198	0.046455	0.037584	0.031376	0.027544	0.051310	0.041598	0.033591	0.028889	0.025918
3/2	0.051203	0.041947	0.034675	0.029735	0.026308	0.056188	0.046431	0.037289	0.030940	0.027320
4/2	0.062356	0.052618	0.043006	0.034908	0.029689	0.066675	0.056933	0.047222	0.038299	0.031793
Hes	0.051252	0.041917	0.034689	0.029861	0.026433	0.054005	0.044285	0.035841	0.030307	0.026837
Scott	0.043995	0.034796	0.029360	0.026487	0.023977	0.051267	0.041565	0.033614	0.028921	0.025925
α-H	0.057053	0.047299	0.038255	0.031794	0.027833	0.048696	0.039384	0.032532	0.028348	0.025379
	Double Exponential Jumps: $T = 0.5, M = 50$									
SS	0.044894	0.035988	0.028514	0.022641	0.018202	0.035143	0.025923	0.018926	0.014571	0.011948
HW	0.039668	0.029942	0.021190	0.015469	0.012436	0.034779	0.025096	0.017311	0.013299	0.011259
3/2	0.034683	0.025483	0.018545	0.014262	0.011734	0.039655	0.029915	0.020841	0.014953	0.012166
4/2	0.045834	0.036091	0.026504	0.018609	0.014006	0.050156	0.040405	0.030706	0.021884	0.015812
Hes	0.034734	0.025470	0.018611	0.014440	0.011898	0.037476	0.027781	0.019523	0.014565	0.011941
Scott	0.027494	0.018330	0.013505	0.011537	0.010117	0.034737	0.025067	0.017355	0.013348	0.011277
α-H	0.040522	0.030784	0.021833	0.015827	0.012646	0.032178	0.022923	0.016478	0.013041	0.011037
	Mixture of Normal Jumps: $T = 0.25, M = 50$									
SS	0.057166	0.047972	0.040270	0.034388	0.030039	0.046266	0.037355	0.031020	0.027271	0.024859
HW	0.053264	0.043322	0.034634	0.028936	0.025962	0.048271	0.038424	0.030804	0.026791	0.024849
3/2	0.048242	0.039627	0.033236	0.028882	0.025858	0.053287	0.043275	0.034396	0.028643	0.025820
4/2	0.059158	0.049215	0.039545	0.031899	0.027438	0.063638	0.053771	0.043881	0.035158	0.029233
Hes	0.048220	0.039072	0.032297	0.028130	0.025455	0.050054	0.040253	0.032374	0.027737	0.025312
Scott	0.040716	0.031905	0.027075	0.025189	0.023608	0.049247	0.039412	0.031614	0.027250	0.025071
α-H	0.054254	0.044268	0.035301	0.029254	0.026131	0.046352	0.037027	0.030351	0.026752	0.024653

Table 17: Variance call options with SV. Parameters from Table 11 and jump parameters from Table 12. For PROJ: Prices found with $N = 2^{10}$, $m_0 = 40$, and $L_1 = 14$.

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A Appendix

A.0.1 Variance Grid Bounds

To determine the grid boundaries v_1 and v_{m_0}, define $\bar{\mu}(t) := \mathbb{E}[v_t|v_0]$ and denote $\bar{\sigma}(t)$ as the standard deviation of v_t conditional on v_0.[22] We fix $t = T/2$ and center the grid about the mean of the variance process v_t by

$$v_1 := \begin{cases} \max\{\bar{v}, \bar{\mu}(t) - \gamma\bar{\sigma}(t)\} & \text{if} \quad v_t \quad \text{is a positive process,} \\ \bar{\mu}(t) - \gamma\bar{\sigma}(t) & \text{if} \quad v_t \quad \text{could takes negative values,} \end{cases} \tag{47}$$

$$v_{m_0} := \bar{\mu}(t) + \gamma\bar{\sigma}(t),$$

where we define the constants $\gamma = 4 \sim 9$ (for realized variance products) and $\bar{v} = 10^{-4}$ for models, which require a positive level of v_t. For the Stein-Stein and 4/2 models, $\bar{v} = 0.01$ is recommended. In the numerical experiments, we fix $\gamma = 5.5$ for all models except $\alpha-$Hypergeometric and Hull-White, for which we set $\gamma = 8.5$. We note that for Heston's model, 3/2 model, and 4/2 model in Table 1,

$$\bar{\mu}(t) = e^{-\eta t}v_0 + \theta(1 - e^{-\eta t}), \qquad \bar{\sigma}^2(t) = \frac{\sigma_v^2}{\eta}v_0(e^{-\eta t} - e^{-2\eta t}) + \frac{\theta\sigma_v^2}{2\eta}(1 - e^{-\eta t} + e^{-2\eta t}), \tag{48}$$

[22] If moments of the variance process are unknown, we can simulate $\bar{\mu}(t)$ and $\bar{\sigma}(t)$ using, for example, the Euler scheme.

where for the reformulated 3/2 model we use $\widehat{\eta}, \widehat{\theta}$ and $\widehat{v}_0$ in equation (48). In the Stein-Stein and Scott models, we have

$$\bar{\mu}(t) = e^{-\eta t} v_0 + \theta(1 - e^{-\eta t}), \qquad \bar{\sigma}^2(t) = \frac{\sigma_v^2}{2\eta}(1 - e^{-2\eta t}). \tag{49}$$

For the Hull-White model, we have

$$\bar{\mu}(t) = v_0 e^{a_v t}, \qquad \bar{\sigma}^2(t) = v_0^2 e^{2a_v t}(e^{\sigma_v^2 t} - 1).$$

Since the mean and variance of v_t are unknown in the α-Hypergeometric model, to estimate $\bar{\mu}(t)$ and $\bar{\sigma}(t)$ we can apply equation (49) with the parameters $\bar{\eta}, \bar{\theta}$, and σ_v of the linear approximation

$$dv_t \approx \theta a e^{av_0}\left(\frac{\eta - \theta e^{av_0}(1 - av_0)}{\theta a e^{av_0}} - v_t\right) dt + \sigma_v dW_t^2 := \bar{\eta}(\bar{\theta} - v_t)dt + \sigma_v dW_t^2.$$

A.0.2 Biorthogonal Projection

Consider a random variable X with an unknown density f_X, for which the characteristic function $\phi_X(\xi)$ is available (or can be approximated). Given a (compactly supported) *generator* $\varphi(x)$, a *resolution* $a > 0$, and a reference point x_1, define the set of basis elements $\varphi_{a,n}(x) := a^{1/2}\varphi(a(x - x_n))$. The orthogonal projection $P_{\mathcal{M}_a} f_X$ of f_X onto $\mathcal{M}_a := \overline{\text{span}}\{\varphi_{a,n}\}_{n\in\mathbb{Z}}$ is determined by a *dual basis* $\{\widetilde{\varphi}_{a,n}\}_{n\in\mathbb{Z}}$ characterized by

$$P_{\mathcal{M}_a} f_X(x) = \sum_{n\in\mathbb{Z}} \langle f_X, \widetilde{\varphi}_{a,n}\rangle \varphi_{a,n}(x),$$

where the dual basis is *biorthogonal* in the sense that $\langle \varphi_{a,k}, \widetilde{\varphi}_{a,n}\rangle = \mathbb{1}_{\{k=n\}}$, and the special case of an orthogonal basis is self-dual. The projection coefficients are obtained in closed-form using the Fourier transform $\widehat{\widetilde{\varphi}}$ of $\widetilde{\varphi}$:

$$\langle f_X, \widetilde{\varphi}_{a,n}\rangle = \frac{a^{-1/2}}{\pi} \Re\left[\int_0^\infty \exp(-\mathrm{i} x_n \xi) \cdot \phi_X(\xi) \widehat{\widetilde{\varphi}}\Big(\frac{\xi}{a}\Big) d\xi\right], \tag{50}$$

given that $\widehat{\widetilde{\varphi}}(\xi)$ is known. In this paper, we utilize the cubic B-spline generator

$$\varphi(y) = \begin{cases} (y+2)^3/6, & y \in [-2,-1] \\ 2/3 - y^3/2 - y^2, & y \in [-1,0] \\ 2/3 + y^3/2 - y^2, & y \in [0,1] \\ (2-y)^3/6, & y \in [1,2]. \end{cases} \tag{51}$$

After fixing a, N and x_1, we restrict $\mathcal{M}_a$ to the finite set $\{x_n\}_{n=1}^N$, where each basis element $\varphi_{a,n}(x)$ is centered over the grid point $x_n = x_1 + (n-1)/a$. Accounting for the Nyquist frequency, we define an N-point frequency grid

$$\Delta_\xi = 2\pi a/N, \qquad \xi_n = (n-1)\Delta_\xi, \quad n = 1, ..., N, \tag{52}$$

which is used to numerically invert the analytical coefficient representation in equation (50). The final approximation is given by

$$f_X(x) \approx a^{1/2} \Upsilon_{a,N} \sum_{1\le k\le N} \bar{\beta}_{a,k} \cdot \varphi_{a,k}(x)$$

with constant $\Upsilon_{a,N} := 32a^4/N$. The coefficients $a^{1/2}\Upsilon_{a,N} \cdot \bar{\beta}_{a,k} \approx \beta_{a,k} = \langle f_X, \widetilde{\varphi}_{a,k} \rangle$ are found using the exponentially convergent discretization

$$\left\{\bar{\beta}_{a,k}\right\}_{k=1}^{N} := \Re\left\{\mathcal{D}\left\{H_j\right\}\right\}, \quad \mathcal{D}_n\left\{H_j\right\} = \sum_{j=1}^{N} e^{-\mathrm{i}\frac{2\pi}{N}(j-1)(n-1)} H_j, \quad n = 1, ..., N, \tag{53}$$

where $\mathcal{D}$ is the discrete Fourier transform (DFT). The DFT input vector $\{H_j\}_{j=1}^{N}$ is defined by

$$H_1 := 1/32a^4, \qquad H_n := \phi_X(\xi_n) \cdot \zeta_n \cdot \exp(-\mathrm{i}\xi_n \cdot x_1), \quad n \geq 2. \tag{54}$$

where

$$\zeta_n := \frac{2520(\sin(\xi_n/(2a))/\xi_n)^4}{1208 + 1191\cos(\xi_n/a) + 120\cos(2\xi_n/a) + \cos(3\xi_n/a)}, \quad n \geq 2. \tag{55}$$

For details on the derivation of $\widehat{\widetilde{\varphi}}$ (and ζ_n) for B-spline bases, see Kirkby (2017b).

Jump Type	c_2^J	c_4^J
Normal	$\lambda(a_1^2 + b_1^2)$	$\lambda(a_1^4 + 6b_1^2a_1^2 + 3b_1^4\lambda)$
DE	$2\lambda\left(\frac{p}{\eta_1^2} + \frac{(1-p)}{\eta_2^2}\right)$	$24\lambda\left(\frac{p}{\eta_1^4} + \frac{1-p}{\eta_2^4}\right)$
Mixed Normal	$p\lambda(a_1^2 + b_1^2) + (1-p)\lambda(a_2^2 + b_2^2)$	$p\lambda(a_1^4 + 6b_1^2a_1^2 + 3b_1^4\lambda) + (1-p)\lambda(a_2^4 + 6b_2^2a_2^2 + 3b_2^4\lambda)$

Table 18: Cumulants of common jump distributions (defined in Table 2).

Remark 8 *To determine the truncated density support parameter α, we consider a simple rule based on cumulants of the log return process in the regime-switching model. Recall from the RS model, we have the set of symbols*

$$\psi_j(\xi) = i\xi\mu_j - \frac{1}{2}\xi^2\sigma_j^2 + \lambda(\phi(\xi) - 1), \quad j = 1, \ldots, m_0, \tag{56}$$

where the jump and diffusion components are independent. Let c_i^J denote the ith cumulant of the jump component, defined in Table 2 for jump distributions considered in this paper. As a variant of the approach in Fang and Oosterlee (2008), we use the grid defined by

$$\bar{\alpha} = \max\left\{\frac{1}{2},\ L_1 \cdot \sqrt{c_2^* T + \sqrt{c_4^* T}}\right\}, \quad c_2^* := c_2^J + m^2(\bar{\mu}(T/2)), \quad c_4^* := c_4^J,$$

where we recommend $L_1 = 10 \sim 14$. The second cumulant of the diffusion component is given by $m(v)$, defined in equation (5), and $\bar{\mu}(t) = \mathbb{E}[v_t|v_0]$ is provided in Section A.0.1.

A.0.3 Gaussian Quadrature

To approximate Ψ, we apply Gaussian quadrature over each interval $I_k = [x_k - 2\Delta, x_k + 2\Delta]$, $k = 1, \ldots, N$. After a change of variables, this is accomplished by applying an N_q-point quadrature to each subinterval of $[-2, 2] = [-2, -1] \cup [-1, 0] \cup [0, 1] \cup [1, 2]$ as follows. Specifically,

$$\begin{aligned}
\Psi(n, k) &:= \Upsilon_{a,N} \cdot a^{1/2} \int_{I_k} \exp(\mathrm{i}\xi_n h(x)) \cdot a^{1/2}\varphi(a(y - x_k))dy \\
&= \Upsilon_{a,N} \int_{[-2,2]} \exp\left(\mathrm{i}\xi_n h\left(x_k + \frac{y}{a}\right)\right)\varphi(y)dy \\
&\approx \Upsilon_{a,N} \sum_{l=1}^{4\cdot N_q} \omega_l \cdot \exp\left(\mathrm{i}\xi_n h\left(x_k + \gamma_l\right)\right)\varphi(\gamma_l),
\end{aligned}$$

where $\{(\gamma_l, \omega_l)\}_{l=1}^{4\cdot N_q}$ are the nodes and weights on $[-2, 2]$. Next define the sample grid

$$\eta_j, \quad j = 1, ..., N_\eta, \qquad N_\eta := ((N-1)+4) \cdot N_q, \tag{57}$$

where the set of η_j is a refined grid over $x_1 - 2\Delta, \ldots, x_N + 2\Delta$ so that each subinterval of length Δ contains N_q points from the set $\{\eta_j\}_{j=1}^{N_\eta}$. There are $(N-1)+4$ such intervals. The Gaussian approximation of $\Psi(n, k)$, for $n = 1, \ldots, N$ and $k = 1, \ldots, N$, is given by

$$\bar{\Psi}(n, k) := \sum_{l=1}^{4\cdot N_q} \theta^n_{N_q(k-1)+l} \cdot \sigma_l = \sum_{l=1}^{2\cdot N_q} \left(\theta^n_{N_q(k-1)+l} + \theta^n_{N_q(k+3)+1-l} \right) \cdot \sigma_l, \tag{58}$$

where

$$\sigma_l := \Upsilon_{a,N} \cdot \varphi(\gamma_l) \cdot \omega_l, \quad l = 1, ..., 2 \cdot N_q,$$

and

$$\theta^n_j := \exp\left(\mathrm{i}\xi_n h(\eta_j)\right), \qquad n = 1, ..., N, \quad j = 1, ..., N_\eta.$$

We apply this in Remark 9 with the $N_q = 5$ point Gaussian quadrature, so each basis element $\varphi_{a,k}(x)$ is integrated against $\exp(i\xi h(x))$ using a total of $4 \cdot N_q = 20$ points.

Remark 9 *To numerically integrate $\Psi(n, k)$, we apply a composite Gaussian quadrature which consists of a five point rule applied over each interval $[r, r+1]$, for $r \in \{-2, -1, 0, 1\}$. By symmetry of $\varphi(y)$, we only need to determine the nodes $\{\gamma_l\}$ and weights $\{\omega_l\}$ for $r \in \{-2, -1\}$ (see equation (58)). On $[-2, -1]$, we obtain*

$$\{\gamma_l\}_{l=1}^5 = \Big\{ -\frac{3}{2} - g_3, -\frac{3}{2} - g_2, -\frac{3}{2}, -\frac{3}{2} + g_2, -\frac{3}{2} + g_3 \Big\}, \quad \{\omega_l\}_{l=1}^5 = \frac{1}{2}\{v_3, v_2, v_1, v_2, v_3\},$$

while on $[-1, 0]$ we have

$$\{\gamma_l\}_{l=6}^{10} = \Big\{ -\frac{1}{2} - g_3, -\frac{1}{2} - g_2, -\frac{1}{2}, -\frac{1}{2} + g_2, -\frac{1}{2} + g_3 \Big\}, \quad \{\omega_l\}_{l=6}^{10} = \frac{1}{2}\{v_3, v_2, v_1, v_2, v_3\},$$

where we define the constants $g_2 := \frac{1}{6}(5 - 2\sqrt{10/7})^{1/2}$, $g_3 := \frac{1}{6}(5 + 2\sqrt{10/7})^{1/2}$, $v_1 := 128/225$, $v_2 := (322 + 13\sqrt{70})/900$, $v_3 := (322 - 13\sqrt{70})/900$. The final set of weights $\sigma_l = \Upsilon_{a,N} \cdot \varphi(\gamma_l) \cdot w_l$ are determined by evaluating the cubic generator $\varphi(y)$ defined in equation (51) at each of $\{\gamma_l\}_{l=1}^{10}$, and we store σ_l for repeated use.